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Simultaneity of certainty in Turkish Sign Language (TİD)

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ABSTRACT

As collaborative discourse participants, interlocutors are expected to convey how certain they are about the proposition that they put forward in the discourse. In spoken language literature, speakers use various strategies to convey their certainty like attitude verbs, prosody, or gestures. Sign language literature reports that signers mostly modulate their manual (hands) signs and nonmanuals (face and body movements), yet there is no study investigating how both channels interact in the expression of certainty. The current study tests the degree to which different sentence types and nonmanuals affect the signer certainty in Turkish Sign Language (TİD) in a rating study. I show that signers interpret signer certainty by taking into account cues from both manual and nonmanual channels. Within the framework of threshold semantics, I discuss how sentences set the certainty threshold, and show how nonmanuals boost or deboost the threshold. Specifically, head nod increases the threshold while squint decreases it, thus providing further support for the dynamic nature of thresholds.

1. Introduction

Interlocutors in conversations assume an implied contract between them, that is, that everyone is telling the truth to have efficient and successful transmission of information (Grice, 1975). Based on this sociolinguistic contract, if the interlocutor is not certain about the proposition that they put forward, she is expected to signal it so as not to be blamed for violating the contract, that is, lying (Krifka, 2017, 2023). While testing how speakers convey their commitment, some scholars operationalized the commitment as the degree of speaker (un)certainty and their knowledge about the truth of the proposition (Lorson et al., 2021). To convey various epistemic strengths, speakers can utilize several mechanisms such as different intonation contours (Heritage, 2013; Heim, 2019; Heim and Wiltschko, 2020), using modals or attitude verbs like *know* or *think* (Lorson et al., 2021), or the interaction of sentence types and intonation (Tonhauser, 2016; Prieto and Roseano, 2021). Furthermore, studies on multimodality also showed that speakers use visual cues along with their prosody in various structures to convey their epistemic stance (Borràs-Comes et al., 2019; Tubau et al., 2015; Swerts and Krahmer, 2005). Multimodal realizations as intonational patterns and facial expressions are found to be interrelated in pragmatic encoding of epistemic stance (Borràs-Comes et al., 2019; Tubau et al., 2015).

Similarly, signers in sign language literature were also reported to use various strategies like modals or attitude verbs (Karabüklü, 2022) or modulating the manual signs as slower and reduplicated movements in attitude verbs for lower certainty levels or sharp and single movements for higher levels (Shaffer, 2004). In addition to the modifications in manual channel, signers were also observed to modulate their nonmanual (facial and body movements) cues. For instance, American Sign Language (ASL) signers decrease or increase head or brow movements in line with the certainty (Shaffer, 2004). Turkish Sign Language (TİD) signers also showed similar patterns by intensifying nonmanuals along with the levels of certainty (Karabüklü and Wilbur, 2020). In addition, Japanese Sign Language (JSL) signers increased the degree of head tilt with lower

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degrees of certainty (Akahori et al., 2013). Similar patterns for various nonmanuals were also reported for German Sign Language (DGS), Sign Language of Netherlands (NGT), and Irish Sign Language (ISL) (Herrmann, 2013), and for Iranian Sign Language (ZEI) (Siyavoshi, 2019).

In spoken languages, the interaction of speech and gestures is well-examined. By examining the temporal alignment of prosody and gestures in encoding of epistemic stance, Borràs-Comes et al. (2019) tested if speakers can associate the epistemic information conveyed in intonation and gesture in matching tasks. During material preparation, they observed that high commitment statements appeared with head nod and beat gestures while low commitment statements appeared with head tilt, eyebrow raise, shoulder shrug, squinted eyelids, and open palms. High commitment questions appeared with repetitive head nod, eyebrow raise, and deictic gestures while low commitment questions appeared with head tilt and eyebrow furrowing. They found that speakers were able to produce the corresponding intonational patterns according to epistemic stance in silent gestures and to associate the intonational input with the silent gestures. Their study clearly showed that two modalities (speech and gesture) were used to convey the same pragmatic notion. Yet, the two channels' (manual and nonmanual) interaction in a single visual modality in sign languages is unknown. To address this gap, this paper examines the following questions: (i) Do manual and nonmanual channels behave similarly to speech and gesture by conveying the same semantic and pragmatic notion? (ii) If they have distinct functions, what are their functions and how do they affect signer certainty?

To answer these questions, I test how signer certainty is affected via different sentence types (manual channel) and different nonmanuals (nonmanual channel) in TİD in a certainty rating study. After presenting the literature on sign languages in Section 2 and on TİD in Section 2.1, I present the current study design to test how certainty levels are affected in Section 3. I then discuss the results in terms of threshold semantics based on the significant interaction results of both channels (manual and nonmanual) in Section 5 and conclude with the crosslinguistic implications of the study in Section 6.

1.1. *Threshold semantics*

When commitment is operationalized as various degrees of certainty, speakers and signers choose expressions conveying the threshold for the probability of the event in the proposition to be true (Yalcin, 2010; Lassiter, 2017). For example, if the expression *know* has a threshold 0.8, then the proposition in (1) is true when the event of Alina's being home exceeds this threshold.[1] If the speaker does not have the best grounds or evidence to utter the sentence in (1), but based on previous knowledge about the situation or about Alina the speaker can guess or infer her whereabouts. As a cooperative interlocutor, the speaker chooses to utter the sentence in (2). In this case, if *think* has a threshold 0.6, then the proposition in (2) is true when the event of Alina's being home exceeds this threshold.

(1) I know Alina is home.
(2) I think Alina is home.

Although the initial analyses put the probabilities as part of the lexical verbs, experimental studies have shown that several factors like speaker identities, social goals, or communicative goals affect the thresholds by showing their dynamic nature. As for individual differences in certainty expressions, English listeners were shown to expect different thresholds from a generic speaker for the same expressions than each other (Schuster and Dengen, 2020). Furthermore, listeners adapted to different thresholds when they were presented with either a cautious speaker or a confident one. Listeners were shown to adapt to their interlocutors' identity (cautious vs. confident) and the thresholds of uncertainty expressions were dynamic (Schuster and Dengen, 2020).

Variation is not only tied to the individuals but also to the situations. Lorson et al. (2021) investigated how English speakers use attitude verbs in different situations like questioning someone and briefing a colleague by putting participants in different power relations. They also asked participants to rate how likely they estimate the event to be. They found that participants used attitude verbs (know, notice, see, believe, be sure, and think) in distinct strength orders in different contexts (briefing and interrogation). Furthermore, in a different study, authors found that speakers' production of *know* and *bare assertion* differ based on whether they are in a cooperative context or not (Lorson et al., 2023).

Politeness is another factor affecting the certainty expressions and thresholds by balancing between communicating (un) certainty and being (im)polite. Speakers conveyed messages in face-threatening conditions with more uncertainty than the ones in non-face-threatening conditions (Holtgraves and Perdew, 2016). Similarly, Catalan Sign Language (LSC) signers were also found to use uncertainty expressions like MY OPINION, THINK, SEE and nonmanuals as shoulder shrug and head tilt to express politeness (Moya-Avilés et al., 2023). Politeness is also discussed to make nonmanuals more subtle or less emphasized in Swiss German Sign Language (DSGS) (Boyas Braem and Tissi, 2023).

Although different expressions have inherent different thresholds, these thresholds are dynamic and affected by various factors. The thresholds can be boosted or deboosted across different speaker identities (cautious or confident) (Schuster and Dengen, 2020), different social goals (debriefing or interrogation) (Lorson et al., 2021, 2023), or communicative motives (politeness) (Holtgraves and Perdew, 2016). Although findings in sign language literature hint that signers also have dynamic semantics for (un)certainty expressions in both manual and nonmanual channels, it is unknown how these channels interact and convey (un)certainty in terms of threshold semantics. To better set the baseline, this study will focus only on the generic

[1] Numeric values are here to exemplify the gradable nature rather than strictly measuring the threshold. Yalcin (2010, p. 923) states the idea as '… the semantics of gradable predicates models them as relations to degrees. Degrees need not correspond to numerical magnitudes, but in the case of predicates that correspond to intuitively measurable properties, they often do.' I follow the perspective here to better understand how manual signs and nonmanuals interact and affect the sentential meaning.

signer by leaving all mentioned factors for future studies. In the current study, I test the effects of modulations in both manual and nonmanual channels on signer certainty in a rating task to investigate how the simultaneous structure of sign languages is organized.

2. Signer certainty in sign languages

Signer certainty has been observed to be encoded with nonmanuals, manual signs, and their modulations like sharp movement or short repetitive movements such as in NGT (Herrmann, 2013), in ASL (Shaffer, 2004), and in JSL (Akahori et al., 2013). Similarly to Borràs-Comes et al. (2019)'s findings in gestures, the patterns reported for ASL and JSL include head movements like head nodding or tilting, and brow movements along with manual modifications. As seen in examples below, when the signer had low certainty the sign SEEM[2] was signed with slow reduplications along with slight brow frown (bf)[3] and head nod (hn) (3). When the signer was neutral about the proposition, there was no modification on the manual sign and nonmanuals (4). In contrast, when the signer certainty increased, the sign SEEM was signed with sharp reduplication, solid brow frown, and sharp head nod. As in alignments of gestures and prosody in Catalan (Borràs-Comes et al., 2019), the patterns observed in ASL also suggest the alignment of the modulations in manual signs and nonmanuals. Furthermore, in terms of threshold semantics, ASL seems to reflect the various thresholds on the certainty onto both manual and nonmanual channels via their modulations. Thus, these modulations should be tested for their robustness across structures.

(3)

[LIBRARY HAVE DEAF LIFE]$_{\text{TOP}}$ $\overline{\text{SEEM}}^{\text{slight bf,hn}}$ slow reduplication
'I think the library may have Deaf life.' (ASL Shaffer 2004: p. 192)

(4)

[LIBRARY HAVE DEAF LIFE]$_{\text{TOP}}$ $\overline{\text{SEEM}}^{\text{bf,hn}}$
'I think the library has Deaf life.' (ASL Shaffer 2004: p. 192)

(5)

[LIBRARY HAVE DEAF LIFE]$_{\text{TOP}}$ $\overline{\text{SEEM}}^{\text{solid bf, sharp hn}}$ sharp reduplication
'Surely, the library has Deaf life.' (ASL Shaffer 2004: p. 192)

A similar finding was also reported for JSL where the degree of head tilt was modified based on the probability in epistemic signs (Fig. 1). Since epistemics are the speaker's perspective on the proposition (Hacquard, 2006), this modification might be also to reflect the signer certainty about the proposition. Rather than manual modifications, authors reported that signers attributed different lexical signs to different degrees (Akahori et al., 2013).

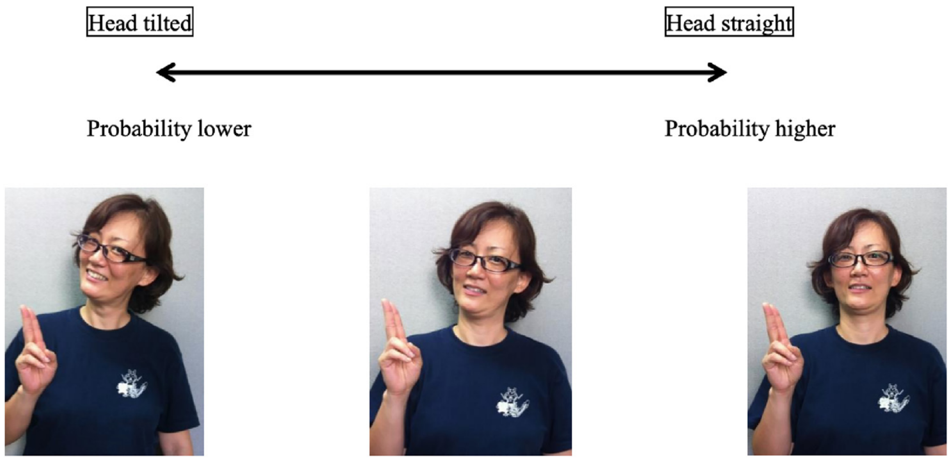

Fig. 1. Modulation of head tilt with probability (Akahori et al., 2013).

[2] Sign languages do not have writing systems, this is why the signs will be glossed with small caps and the nonmanuals will be shown with a line above the signs and abbreviations of observed nonmanuals. The line represents the domain of the nonmanuals.

[3] Abbreviations used throughout the manuscript are: bf - brow frown, br - brow raise, bl - body lean, hn - head nod, ht - head tilt, eo - eyes open wide, sq - squint, lp - lips pursed, rhn - repetitive head nod, htsq - head tilt and squint, rhnsq - repetitive head nod and squint, non - no nonmanual, ASL - American Sign Language, DGS - German Sign Language, DSGS - Swiss German Sign Language, ISL - Irish Sign Language, JSL - Japanese Sign Language, NGT - Sign Language of Netherlands, LSC - Catalan Sign Language, TİD - Turkish Sign Language, and ZEI - Iranian Sign Language.

Crosslinguistic comparisons of certainty expressions come from DGS, NGT, and ISL in the nonmanual domain (Table 1 adapted from Herrmann 2013, p. 153 & 164). The crosslinguistic comparison with the same elicitation methods shows that signers use similar strategies. To illustrate, uncertainty was expressed via slow head nod (hn), brow frown (bf), or brow raise (br) in DGS. Brow frown and brow raise also appeared in NGT and head nod (hn), tilt (ht), and body lean (bl) appeared in ISL (Table 1). Furthermore, similarly to ASL, nonmanuals became intensified (i.e., tenser and bigger) in all three sign languages when the strength of the utterance was increased.

Table 1
Changes in nonmanuals based on the change in the signer certainty in DGS, NGT, and ISL (Herrmann, 2013).

Signer Certainty	NMMs in DGS	NMMS in NGT	NMMs in ISL
Uncertainty and degree of commitment	hn, slow hn, bf, br	bf, br, sq, hn, eo	hn, ht, bl
Strengthening of the utterance	tense, large hf, intensifying NMMs	tense ht, hf, br, eo	tense large ht, hf, br, eo
Softening of the utterance	bf, hn	hn, hf, lp, bf, br	hn, hf

Studies in sign language literature focus largely on modulations in nonmanuals (Akahori et al., 2013; Herrmann, 2013; Shaffer, 2004) and with some focus on modulations in manual signs (Shaffer, 2004). Thus, how signer certainty is expressed seems to affect both domains across sign languages. The next section shows that TİD is no exception to these observations.

2.1. Signer certainty affects manuals and nonmanuals in TİD

Similarly to ASL, NGT, and JSL, TİD exhibits patterns within modulations in the manual signs and intensifications in nonmanuals (Karabüklü and Wilbur, 2020). The degree of signer certainty was found to be reflected via the intensity of nonmanuals and the modulations in manual signs. Authors manipulated the degree of certainty in the given contexts (6–8) and asked the participant to sign the same target sentence in each context where the signer was not instructed for nonmanuals. All modal signs including OLABILIR[4] (possible - epistemic), SERBEST (free - permission), OLUMLU (positive - ability), YAP (do - ability), LAZIM (necessary - deontic), and MECBUR (obligatory - deontic) (Fig. 2) were tested for potential nonmanuals and manual modifications.

(6) Context: There was chocolate in the lab. The next day when you want to have some chocolate, you realized that someone ate it all up. You, Semra, Asli and Suleyman are working in the lab and you don't know who ate it. You're guessing:
Target Sentence: SEMRA CHOCOLATE EAT OLABILIR
'Semra may have eaten the chocolate.'

(7) Context: There was chocolate in the lab. The next day, when you want to have some chocolate, you realized that someone ate all it up. You know that Semra is so fond of chocolate. You're guessing:
Target Sentence: SEMRA CHOCOLATE EAT OLABILIR
'Semra may have eaten the chocolate.'

(8) Context: There was chocolate in the lab. The next day, when you want to have some chocolate, you realized that someone ate all it up. Only you and Semra are working in the lab. You're guessing:
Target Sentence: SEMRA CHOCOLATE EAT OLABILIR
'Semra may have eaten the chocolate.'

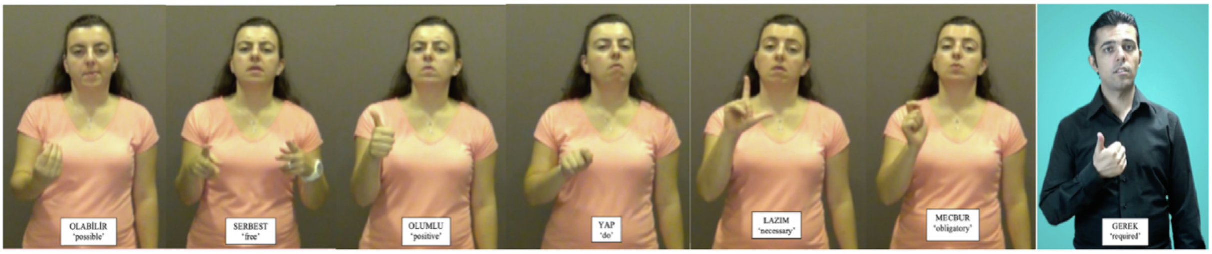

Fig. 2. Modal signs in TİD.

When the target sentences (6–8) were examined, authors reported modulations in manual signs similar to those in ASL. Signs like YAP (do) were observed to have longer path movements when the signer was more certain. This also aligns with the sharper head nod (Karabüklü and Wilbur, 2020). When the second and the last frames of Fig. 3 are compared the sign YAP ends in a lower signing space in the last frame than the second one. The starting points of both the sign and nonmanual are approximately same (first and third frames), yet both manual and nonmanual align at their ending point to reflect the signer certainty. Signs like LAZIM (necessary) that do not have a path movement were observed to be repeated to align with the nonmanuals.

[4] I will gloss the target signs in their native glosses to prevent meaning loss in translation because one sign can be translated into more than one modal or vice versa.

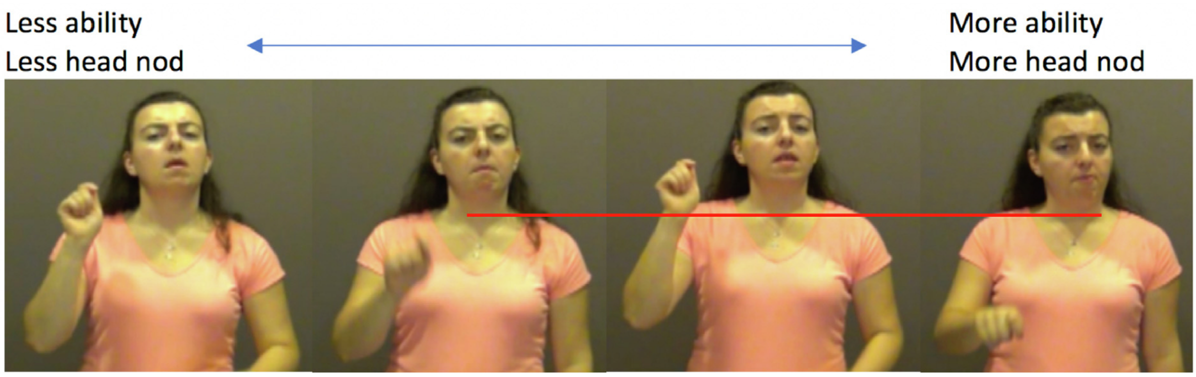

Fig. 3. Intensification of head nod and modulations in the sign YAP(do) (Karabüklü, 2022).

One explanation for the modulations in manual signs is that these signs are focus marked and high certainty is the effect of focus appearing on the modal. That is, speakers are shown to produce focus intonation on the verbal domain for higher certainty levels (Prieto and Roseano, 2021). Considering that focus is marked with increased duration in TİD (Karabüklü and Gürer, 2024), repetitions and longer paths in signs can be for longer duration, and thus for focus. Regarding these patterns, certainty in TİD is marked in ways that are similar to other sign languages and spoken languages.

In addition to the modulations of manuals, nonmanuals showed two patterns: (i) head nod, brow raise, and open eyes were intensified when the signer was more certain about the proposition, and (ii) eye squint was intensified when the signer was more uncertain. As indicated by the red line in Fig. 3, the end of head nod is deeper in the last frame when the signer is more certain about the subject's abilities compared to the second frame where the signer was not as certain. One observation for head nod was its occurrence with the manual sign YES (Açan Aydın, 2014), which marks the positive polarity as seen in sentence (9). Although head nod was not observed in presentational and contrastive focus (Karabüklü and Gürer, 2024), it might be marking polarity or verum, thus yielding higher certainty.

(9)
hn   hn   hn
YES MUSIC/SONG VIOLIN PLAY CHILD STUDENT existential
'Yes, there are school children who are playing music and violins.'
(TİD Açan Aydın 2014: p. 88)

Interestingly, squint shows a negative relation with certainty, that is, it was intensified when the signer was less certain about the proposition. As seen in the first frame in Fig. 4, the squint was at its maximum realization when the signer's eyes were nearly closed when the signer was uncertain about the proposition (6). Then, the squint was visible in the second frame where the signer had more information about the proposition (7). Lastly, the signer barely squinted her eyes in the last frame where the signer certainty was increased (8).

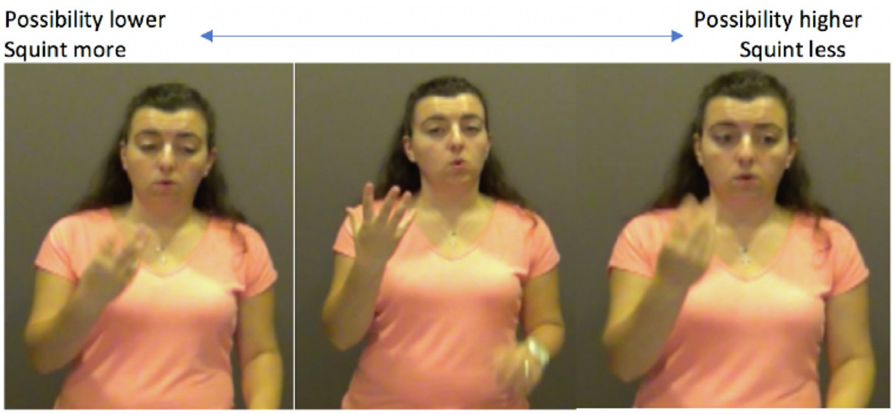

Fig. 4. Intensification of squint with the sign OLABILIR (possible) (Karabüklü and Wilbur, 2020).

Similarly to other sign languages, TİD conveys signer certainty in both manual and nonmanual channels. As seen in spoken languages, modulations in manual signs may be due to information structure. As for nonmanuals, when we examine which nonmanuals appear and their patterns across sign languages, similar nonmanuals appear such as head nod, head tilt, brow movements, or squint. This leads to the following questions: Do these nonmanuals specifically convey the certainty, in other words, are they (un)certainty markers? Or, do nonmanuals have distinct functions and is the certainty effect the interaction of sentences and nonmanuals? How is certainty semantically and pragmatically composed? To answer these questions, we need to look at their distribution in the data and their interaction with each other to answer these questions.

2.2. Sharp head nod and eye squint do not cooccur

To understand the role of nonmanuals, I will focus on their distribution and interaction in Karabüklü and Wilbur's (2020) data. As shown in Table 2, nonmanuals are not a lexical part of modal signs as they can appear with more than one of them. Pointing to this non-lexical proposal, modal signs were also reported with no nonmanuals in natural data (Karabüklü et al., 2018). Interesting two patterns raise from their distributions: (i) Head nod is the most frequently appearing nonmanual with a proportion at least 80% in the data. (ii) Squint is the most consistent nonmanual with a proportion of 70% with epistemic sign OLABILIR (possible). Epistemics are also well-known to convey speaker or signer's justification for the degree of their commitment to the proposition, especially lower level of commitment (Krifka, 2023). It is not unexpected to have the uncertainty effect of squint if it is tied to the epistemic sign. Interestingly, the epistemic sign OLABILIR also had a higher proportion of head nod (90%) that yields high certainty. Although having both squint (uncertainty) and head nod (certainty) with the same sign seems contradictory, these patterns suggest that the threshold of OLABILIR in a certainty scale can be modulated via nonmanuals and that head nod and squint may yield relative (un)certainty to the base threshold.

Table 2
Proportions of nonmanuals in Karabüklü and Wilbur (2020).

Modal Sign/NMM	Squint	Head nod	Brow raise	Head tilt
SERBEST (free - permission)		86%		7%
OLUMLU (positive - ability)	7%	81%	3%	
OLABILIR (possible - epistemic)	70%	90%	85%	35%
LAZIM (necessary - deontic)	28%	97%	89%	38%
YAP (do - ability)	15%	90%	25%	15%

Karabüklü (2022) data was further examined for the occurrences of head nod and squint to examine their interaction (Karabüklü and Wilbur, 2020). Patterns seen in Figs. 3 and 4 are also quantitatively observable with slight or repetitive head nod appearing mostly in uncertain (51%) and slightly uncertain (49%) contexts (Table 3). The occurrence of sharp head nod increases when the certainty increases. The pattern is slightly clearer for squint which decreases across contexts with uncertainty (Table 4). Bare squint appears in slightly uncertain conditions (14%) and decreases in certain contexts. Clear squint shows a steady decrease by appearing most in uncertain contexts (15%) then least in certain contexts (3%).

Table 3
Percentages of head nod occurrence across different signer certainty levels*.

Realization of head nod	Uncertain	Slightly uncertain	Slightly certain	Certain
Slight/repetitive head nod	51%	49%	37%	7%
Clear head nod	34%	29%	30%	34%
Sharp head nod	3%	12%	13%	16%

*Percentages are reported based on Figure 2.10 in Karabüklü (2022: 78).

Table 4
Percentages of squint occurrence across different signer certainty levels*.

Realization of squint	Uncertain	Slightly uncertain	Slightly certain	Certain
Bare squint	7%	14%	9%	11%
Clear squint	15%	10%	9%	3%

*Percentages are reported based on Figure 2.7 in Karabüklü (2022: 74).

The distinct associations of certainty with head nod and squint becomes clearer when their cooccurrences are examined (Table 5). Squint most frequently appeared (63%–38%) when there was no head nod in the data and it disappeared (0%) when there was sharp head nod. Based on their patterns, Karabüklü (2022) argues that sharp head nod signals higher signer certainty while squint is a marker of uncertainty, thus their cooccurrence would yield a contradiction.

Table 5
Percentages of cooccurrences of different realizations of head nod and squint*.

Squint/Head nod	No head nod	Slight/repetitive head nod	Clear head nod	Sharp head nod
Bare squint	63%	13%	10%	0%
Clear squint	38%	12%	16%	0%

*Percentages are reported based on Figure 2.11 in Karabüklü (2022: 79).

As with other languages, TİD clearly uses its simultaneous nature to convey the signer certainty through integrating manual and nonmanual channels. However, it is not clear how two channels interact in TİD and other sign languages. For example, do they set the threshold in the certainty scale separately? Based on the patterns in the tables, does one set the basis like *possible*, and does the other one modify it like scalar modifiers as in *slightly possible*? Furthermore, previous studies suggest that nonmanuals can have distinct effects because they usually function as distinct morphemes (Karabüklü, 2022; Karabüklü and Wilbur, 2020). If nonmanuals behave as morphemes we would expect them to occur with other structures than modals as long as their morpho-semantics allow. Although Karabüklü and Wilbur's (2020) study suggests that nature, they did not test this prediction for nonmanuals with different sentences. In this paper, I explore whether a threshold-based semantics analysis can capture the interaction of manual and nonmanual channels in signer certainty.

3. Methodology

The current study investigates how manual and nonmanual cues interact in the encoding of signer certainty. The study includes three tasks: certainty rating, naturalness rating, and certainty attribution tasks. More specifically, I explore how signer certainty is affected by sentence types (declarative, modals, attitude verbs) and nonmanuals (head nod, squint, head tilt) in a rating study in TİD. Based on the threshold semantics, different sentences as declarative (bare assertion) or attitude reports (know or think) yield different degrees of certainty (Lassiter, 2017; Lorson et al., 2021). Therefore, this study specifically focuses on different sentences in the manual domain rather than the manual modulations reported in the literature. Furthermore, using different sentences with nonmanuals also allows me to test if nonmanuals can occur with various structures and yield similar certainty effects.

To replicate and extend Karabüklü and Wilbur's (2020) findings, different nonmanuals (head nod, head tilt, squint) are tested in combination of all sentences in the stimuli. Since nonmanuals are manipulated for their single occurrence across sentences, the study also includes a naturalness rating task. Thus, we can also test if the naturalness of stimuli affects certainty ratings. The naturalness rating task will show how likely different combinations of sentences and nonmanuals are. Lastly, since sentences include attitude verbs and they are argued to be the reports of the subject (Hintikka, 1969), the study includes a certainty attribution task. To illustrate, the verb *realize* in a sentence (10) reports Maya's attitude to the proposition ‘he made a mistake’ rather than the speakers. In contrast, the speaker's attitude in (10) comes from the adverb *luckily*. Furthermore, the certainty attribution task also shows if signer certainty is due to specific nonmanual cues in sentences with attitude verbs.

(10) Luckily, Maya realized that he made a mistake.

Overall, the study investigates if signer certainty is tied to specific cues in manual and nonmanual channels or if it is composed via these cues together. Furthermore, it also shows how two domains interact in terms of threshold semantics.

3.1. Stimuli

In order to capture the effect of each nonmanual marker, a single sentence was recorded with the following conditions: no nonmanual markers (no nmm) to set the baseline, only one nonmanual marker including squint (sq), head nod (hn), repetitive head nod (rhn), and side to side head tilt (ht), and their combinations including squint and repetitive head nod, and squint and head tilt. Following Karabüklü and Wilbur's (2020) findings on sharp head nod and squint patterns, stimuli did not include their combination. To test the effects of manual channel, stimuli included different sentence types like declarative (bare assertion), modal signs including GEREK (required), LAZIM (necessary), MECBUR (obligatory), OLABILIR (possible), OL LAZIM (be necessary), OLUMLU (positive), SERBEST (free), YAP (do), and attitude verbs including TAHMIN (guess, think), BIL (know), and SOYLE (tell). These attitude verbs were specifically chosen since TAHMIN (guess, think) was reported to occur with squint (Göksel and Kelepir, 2016). In contrast, attitude verbs BIL (know) and SOYLE (tell) were reported to not necessarily have a specific nonmanual (Göksel and Kelepir, 2016). Thus, the stimuli allows us to investigate if these verbs can have head movements and squint to convey signer certainty.

All stimuli had eventive verbs to avoid the ambiguity between stative and eventive interpretations of modals that effect certainty (Ramchand, 2014) and third-person singular subjects to avoid the conflation of the subject and the signer (speaker) (Hacquard, 2006). Another reason to choose third-person subjects was to test whether nonmanuals convey signer's stance on attitude reports like *luckily* in sentence (10).

Since nonmanuals usually cooccur in natural data (11), manipulating their occurrence may yield unnatural results and affect the certainty ratings. To check this possible effect, stimuli also included sentences where target nonmanuals cooccur from Karabüklü and Wilbur's (2020) study as in (11).

(11)
sq,ht
HOUSE $\overline{\text{GO OLABILIR}}$
‘She might have gone to home.’

Stimuli were recorded with a Deaf of Deaf (DoD) consultant and checked for naturalness. Items found too unnatural were rerecorded. Stimuli also included attention checks like 'Please choose 4'.

3.2. Participants

17 participants were recruited; 16 (female: $N = 9$, male: $N = 7$, age $M = 35.19$, $SD = 7.55$, age range = 28–55 years) completed the study. All participants are active members of the Deaf community and self-reported that they use TİD daily. All self-reported 50–75% hearing loss. Seven were Deaf of Hearing (DoH) who were born to hearing parents, that is, they are the first generation of Deaf in their family. They reported to be first exposed to TİD at primary school from their peers at the age of 7. Nine participants were Deaf of Deaf (DoD) who were exposed to TİD from birth. Five of them were second generation Deaf and four were fourth generation Deaf.

Participant pool consists of both DoD and DoH participants reflecting most sign language communities where most signers have DoH background (Zorzi et al., 2022). Sign language literature has shown the effects of age of acquisition (early exposure vs. delayed exposure) of the first language on linguistics skills (Lillo-Martin et al., 2020; Mayberry and Eichen, 1991; Mayberry et al., 2002), academic life (Hrastinski and Wilbur, 2016; Wilbur, 2000), and social life (Pfau et al., 2021). A majority of Deaf children are not exposed to sign language from birth since they are born to hearing parents and do not get early intervention for sign language. Deaf children in that situation are usually exposed to sign language later in life (Mayberry, 2007; Mayberry and Kluender, 2018). Since participants in the current study have different ages of acquisition backgrounds, the possible effects of age of acquisition on the results will be also explored. Yet, since the participant pool is small and may yield null results, the age of acquisition effects need to be tested with a bigger participant pool in future studies.

All participants saw the instructions in TİD and the consent form was explained by the Deaf consultant. All participants signed consent forms stating that their data could be used for academic and scientific purposes. Each participant was paid 10$ per hour for their participation.

3.3. Design

The study had a 12 × 7 mixed effects factorial design where all nonmanual conditions including no nonmanual, squint, head nod, repetitive head nod, head tilt, and their combinations were fully crossed with sentence types declarative, modal signs (OLABILIR, LAZIM, OL LAZIM, MECBUR, GEREK, SERBEST, OLUMLU and YAP), and attitude verbs (TAHMIN (guess, think), BIL (know), and SOYLE (tell)). Two sets were prepared, and participants were randomly assigned to one set. All stimuli are included in the Supplementary Material.

3.4. Procedure

In each session, participants first did a trial to gain familiarity with the task and the scale. They had the chance to take the trial as many times as they wanted. They were told to imagine that they were at a party and saw two people signing. They were then told that they were going to see the signing of one person and asked to use a 7-point slider to indicate how certain the person was about what they were saying. One denoted that the signer was not certain at all while seven denoted that she was fully certain. They were instructed to adjust their ratings based on the certainty level that they thought the signer was (Fig. 5). Slider gave participants the chance to rate the stimuli with in-between values like 3.5 or 5.7 to better represent certainty level than an ordinal rating in a Likert scale.

As mentioned above, manipulating nonmanuals could affect the naturalness, and thus impact the ratings. Thus, participants were also asked to rate each stimulus for their naturalness on a 7-point slider. Naturalness was chosen as a term 'doğal' (natural) in the task since TİD signers uses it to refer to native-like fluent signing.

The last task in the study was a force-choice of certainty attribution, that is, who was certain about the proposition. The options included the signer, the subject, or the embedded subject in the attitude verbs. The participants also had the chance to type their answers if they thought certainty could be attributed to both the signer and the subject. Each participant was randomly assigned to one set and saw the items in a randomized order in Qualtrics. It took participants an hour on average to complete the study ($M = 61.13$, $SD = 16.14$, range = 35–95 min).

3.5. Data coding

Certainty and naturalness ratings, and age were continuous. Certainty attribution data was coded as binary, that is, if the signer was chosen, it was coded as 1, otherwise as 0, likewise for the subject and the embedded subject. Age of acquisition was coded categorically as DoD and DoH. Sentences were coded categorically as declarative, BIL, SOYLE, TAHMIN, OLABILIR, OL LAZIM, GEREK, LAZIM, MECBUR, SERBEST, OLUMLU and YAP. Nonmanuals were also coded categorically as squint, head nod, repetitive head nod, head tilt, repetitive head nod and squint, and head tilt and squint. Due to the internet connection, some participants missed one

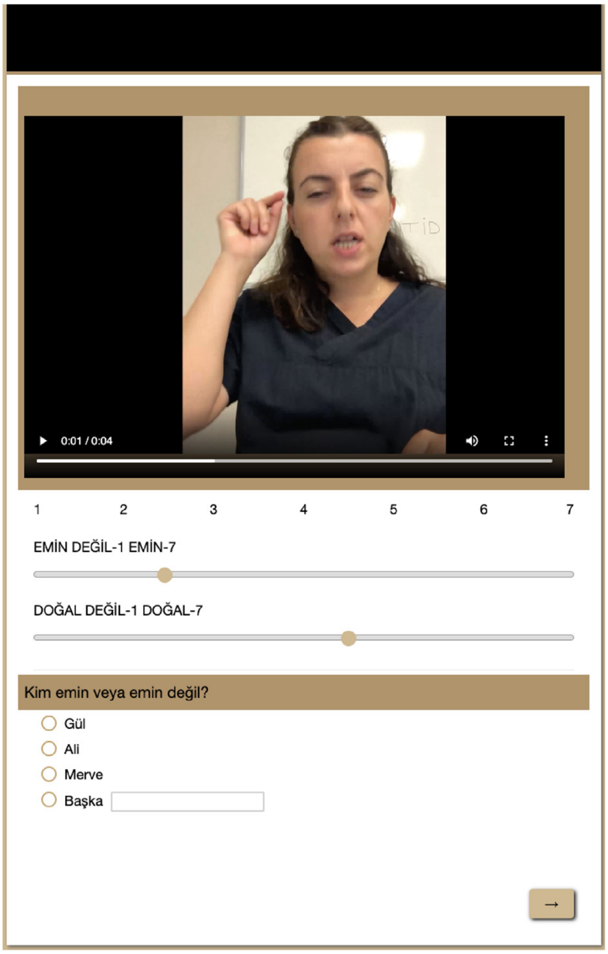

Fig. 5. Snapshot of the rating task.

random item and those were coded as missing items. One participant answered the attention check question incorrectly so that data was excluded. After the exclusion, there were 1372 total data points.

3.6. Data analyses

The design of the experiment included the between-subjects variable AoA (with the two groups: DoD and DoH) and the within-subjects variables nonmanuals and sentence types. All datapoints were analyzed in R using linear models (package *lme4*, Bates et al., 2014). Model comparisons starting with the baseline model were used to derive the significance of the mixed model. The baseline model included random effects and significant confounding factors. Significance levels were derived by comparing models and by using *emmeans* package with Tukey's HSD (Lenth, 2019) in R.

Mixed effects models were used to test the effect of nonmanuals (NMMs), sentence types, and AoA on the ratings of naturalness and certainty, and the choices for the certainty attribution. Mixed effects models also allow random effects on the response variables due to the individual trials and the participants. Thus, Item and Participant were modeled as random factors in the analysis. Including random intercept for the items can explain possible variability that some items might yield more or less certainty than others. Additionally, participants may be prone to distinguish certainty levels differently as some may make more fine-grained distinctions. Thus, random intercept for the participants was also considered. Lastly, some combinations of nonmanuals and sentences in the items might yield more or less certainty for some participants but not for others. Hence, the random slopes were also considered for the nonmanuals and sentences by participants.

Treatment contrast was used for the factors with two levels: age of acquisition (child of deaf parents (DoD) = 0, child of hearing parents (DoH) = 1). I made use of the utility provided by lme4, which creates “n-1” dummy variables for factors with three or more levels by default for the factors: sentences (bil (know) = 0, and dummy variables for declarative, olabilir (possible), lazim (necessary), gerek (required), mecbur (obligatory), ol lazim (be necessary), olumlu (positive), yap (do), serbest (free), soyle (tell), and tahmin (guess)), and nonmanuals (head nod = 0, and dummy variables for head tilt, repetitive head nod, squint, head tilt and squint, repetitive head nod and squint, and no nonmanual), and whocertain (embedded subject = 0, and dummy variables for subject and signer).

A series of models was run to account for possible confounding fixed factors such as gender and age. These are only mentioned in the rest of the manuscript if they are significant. I used a step-by-step approach to establish a model that best explains the data, adding a factor or an interaction term on each step, while assessing model performance with likelihood-ratio tests (Barr et al., 2013). Differences in the χ^2 values, degrees of freedom from the compared models, and the p-values from the likelihood-ratio tests are reported. The final baseline model was determined based on the best fit based on ANOVA tests or the model with the maximal random effects that converged (Barr et al., 2013). All stimuli, data, descriptive statistics, and analysis scripts are presented in the Supplementary Materials.

4. Results

4.1. Naturalness ratings

A linear mixed effects model was fit to assess the effects of sentences, nonmanuals, and age of acquisition[5] on the naturalness ratings. The effects were compared to the baseline model which included the random effect of item and participant. A series of models was also run to test the effects of confounding factors like gender and age.

Results showed significant main effects of sentences ($X^2(11) = 68.617$, $p < 0.001$) and nonmanuals ($X^2(6) = 110.95$, $p < 0.001$). There were a significant interaction effect of sentences and nonmanuals ($X^2(66) = 107.47$, $p < 0.001$) and a significant interaction effect of nonmanuals and age of acquisition ($X^2(7) = 21.727$, $p = 0.003$) on naturalness ratings. Table 6 summarizes the model.

Table 6
Naturalness model summary and fit statistics.

Random effects	**Variance**		**SD**	
Item	0.07		0.26	
Participant	0.84		0.92	
Number of obs: 1372	Groups: Trial = 176		Participant = 15	
Fixed Effects	***df***	***F***	***p***	
Sentence	11	68.617	<0.001	
NMM	6	110.95	<0.001	
Sentence*NMM	66	107.47	<0.001	
NMM*AoA	7	21.727	0.003	
Deviance (REML)	**log Lik.**	***df***	**AIC**	**BIC**
5434.2	−2717	1278	5662.2	6113.3

4.1.1. Effects of sentence types

As seen in Fig. 6, estimated marginal means comparisons showed that GEREK was rated significantly less natural than all other sentence conditions ($p < 0.001$). All contrasts can be found in the Supplementary Material. Naturalness ratings of GEREK reflects the findings that TİD signers consider it as non-native lexical item that is borrowed from Turkish (Karabüklü, 2022). That is why the participants found the stimuli with GEREK as the least natural.

LAZIM was rated significantly less natural than OLABILIR ($\beta = -1.20$, $SE = 0.33$, $t = -3.64$, $p = 0.02$) and OLUMLU ($\beta = -1.75$, $SE = 0.35$, $t = -5.02$, $p < 0.001$). MECBUR was also rated significantly less natural than OLABILIR ($\beta = -1.30$, $SE = 0.33$, $t = -3.91$, $p = 0.01$) and OLUMLU ($\beta = -1.84$, $SE = 0.35$, $t = -5.28$, $p < 0.001$). OL LAZIM was found significantly less natural than OLABILIR ($\beta = 1.10$, $SE = 0.33$, $t = 3.33$, $p = 0.04$) and OLUMLU ($\beta = -1.63$, $SE = 0.35$, $t = -4.73$, $p < 0.001$). Lastly, OLUMLU was rated significantly more natural than TAHMIN ($\beta = 1.20$, $SE = 0.36$, $t = 3.30$, $p = 0.05$).

Multiple explanations could account for the significant differences among other conditions. One is that the contrasts of sentences are averaged over the other variables like nonmanuals and age of acquisition, thus differences could be because some sentences are more natural with specific nonmanuals than others. For example, TAHMIN naturally occurs with more squint (Göksel and Kelepir, 2016) than sharp head nod while YAP naturally occurs with sharp head nod (Özsoy et al., 2018). Overall, all sentences except the ones with GEREK were rated as natural in the range of 4.01 and 5.85.

[5] As noted in Section 3.2, the numbers of participants for age of acquisition groups are small, and the results are reported as explanatory findings and needs to be re-tested with bigger participant pool.

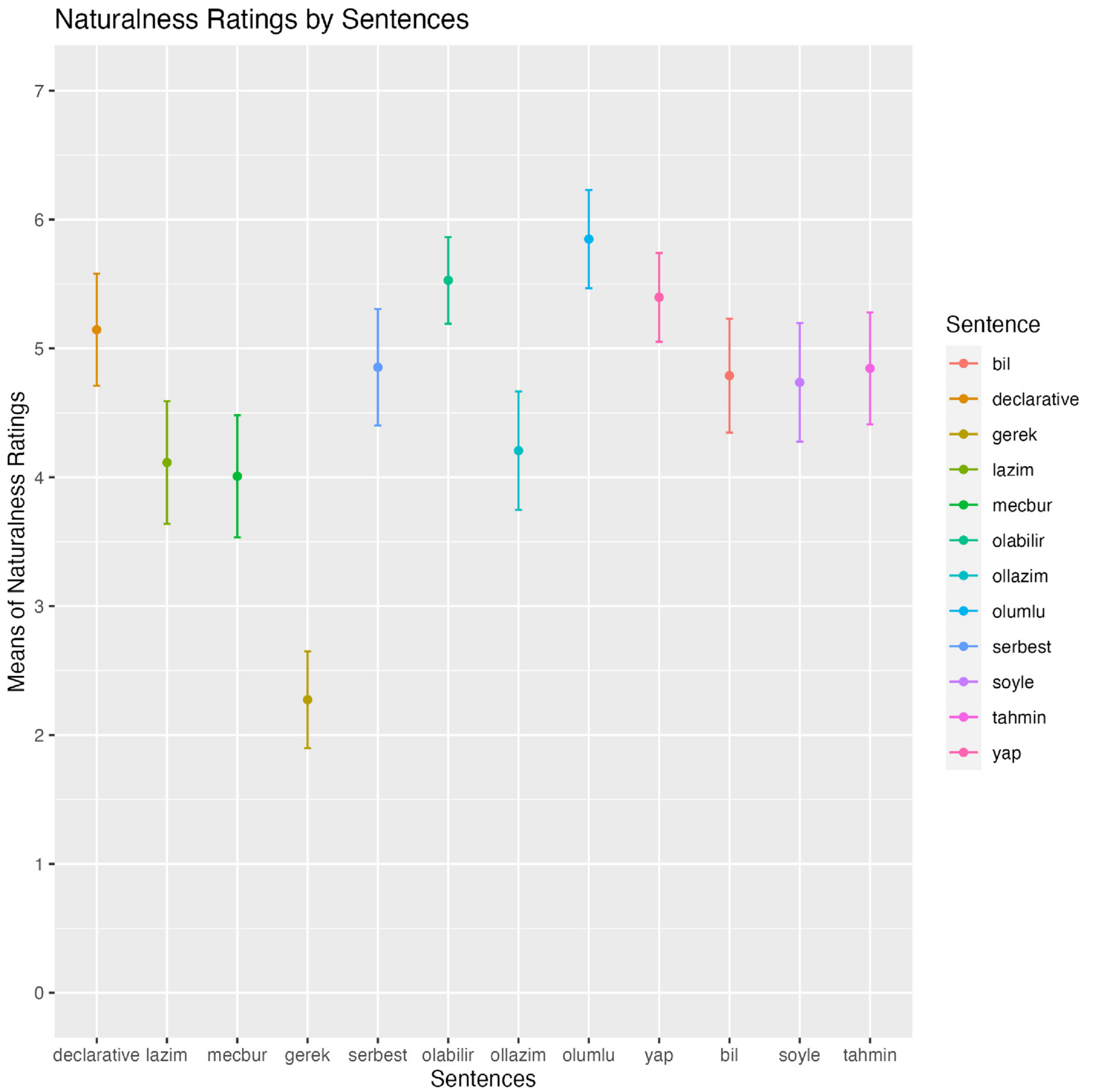

Fig. 6. Effects of nonmanuals on naturalness ratings.

4.1.2. Effects of nonmanuals

As for nonmanuals, estimated marginal means comparisons showed that head tilt was rated significantly less natural than all other conditions ($p < 0.001$, Fig. 7). In contrast to head tilt, head nod and repetitive head nod were rated as the most natural nonmanuals with the given sentences ($p < 0.001$). One reason for head tilt's unnatural rating may be due to its single appearance over the sentences. As seen in comparisons (Fig. 7), it was rated more natural when it cooccurred with head tilt. Furthermore, head nod and squint were reported as single nonmanual in a tested structure like attitude verbs (Göksel and Kelepir, 2016), modals (Karabüklü et al., 2018; Özsoy et al., 2018). Head tilt has not been reported alone as being part of a structure, but instead, they are reported together with other nonmanuals (Karabüklü et al., 2018; Karabüklü and Wilbur, 2020). Hence, head tilt may be a secondary cue along with other nonmanuals and its single occurrence is apparently less natural. Additionally, its domain and timing over a structure might be shifted without other nonmanuals, thus yielding less natural results.

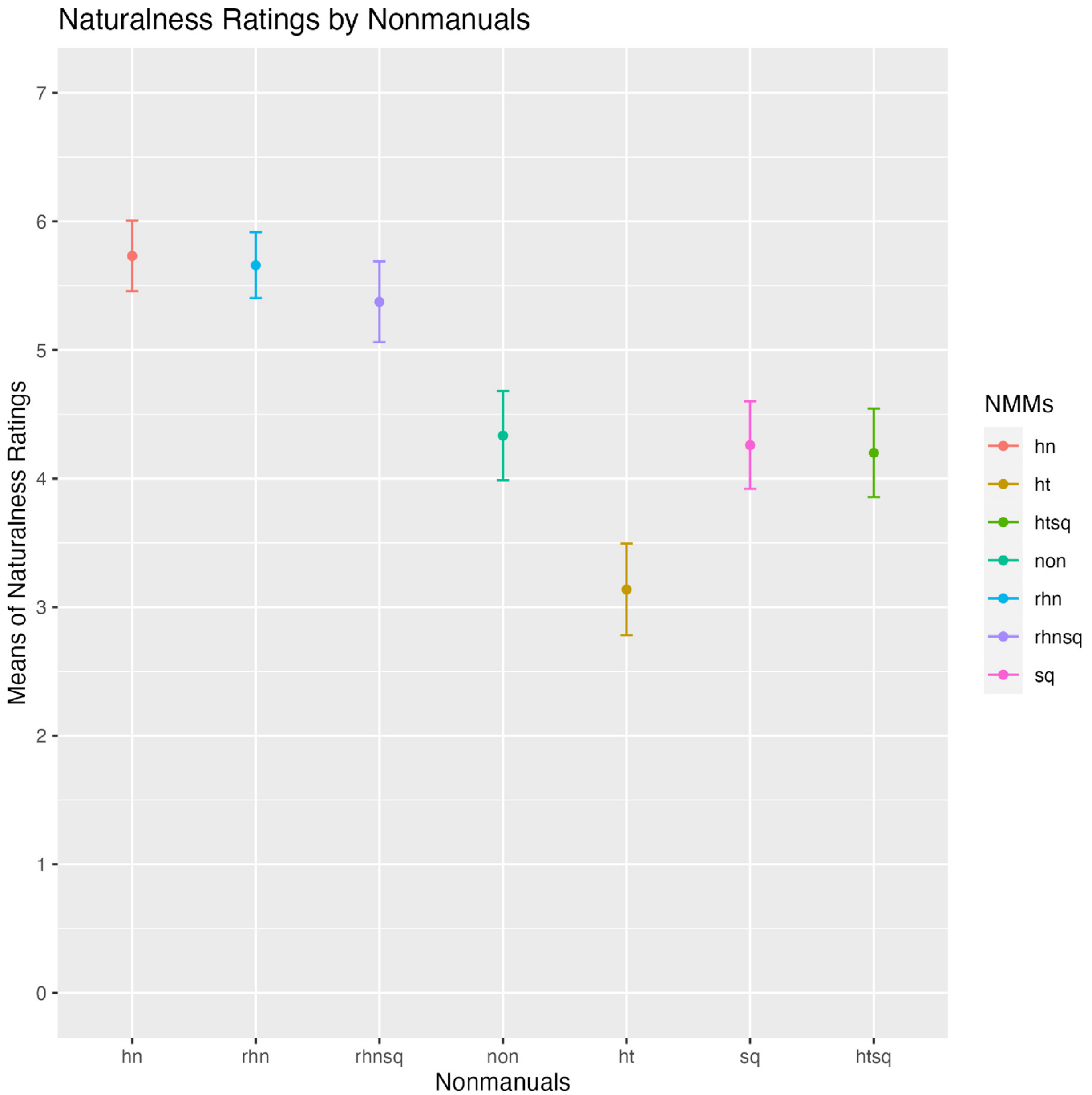

Fig. 7. Effects of nonmanuals on naturalness ratings.

When the interaction effects of age of acquisition and nonmanuals are examined (Fig. 8), DoD participants found no nonmanual (non) condition significantly less acceptable than DoH participants ($\beta = -1.36$, $SE = 0.57$, $t = -2.40$, $p = 0.02$). Specifically, DoD participants found no nonmanual conditions significantly less natural than head nod ($\beta = -1.94$, $SE = 0.31$, $t = -6.17$, $p < 0.001$), repetitive head nod ($\beta = -1.69$, $SE = 0.31$, $t = -5.54$, $p < 0.001$), and repetitive head nod and squint ($\beta = -1.31$, $SE = 0.32$, $t = -4.08$, $p = 0.001$), yet DoH participants did not make a significant distinction between these categories. In contrast, DoH participants rated no nonmanual condition significantly more acceptable than head tilt ($\beta = 1.48$, $SE = 0.33$, $t = 4.44$, $p = 0.0002$), and head tilt and squint ($\beta = 1.21$, $SE = 0.33$, $t = 3.69$, $p = 0.004$) while DoD participants did not make a significant distinction. Lastly, DoD participants also found head tilt significantly less natural than head tilt and squint ($\beta = -1.15$, $SE = 0.32$, $t = -3.59$, $p = 0.01$), and squint ($\beta = -1.19$, $SE = 0.32$, $t = -3.73$, $p = 0.004$) while DoH participants did not make any significant difference.

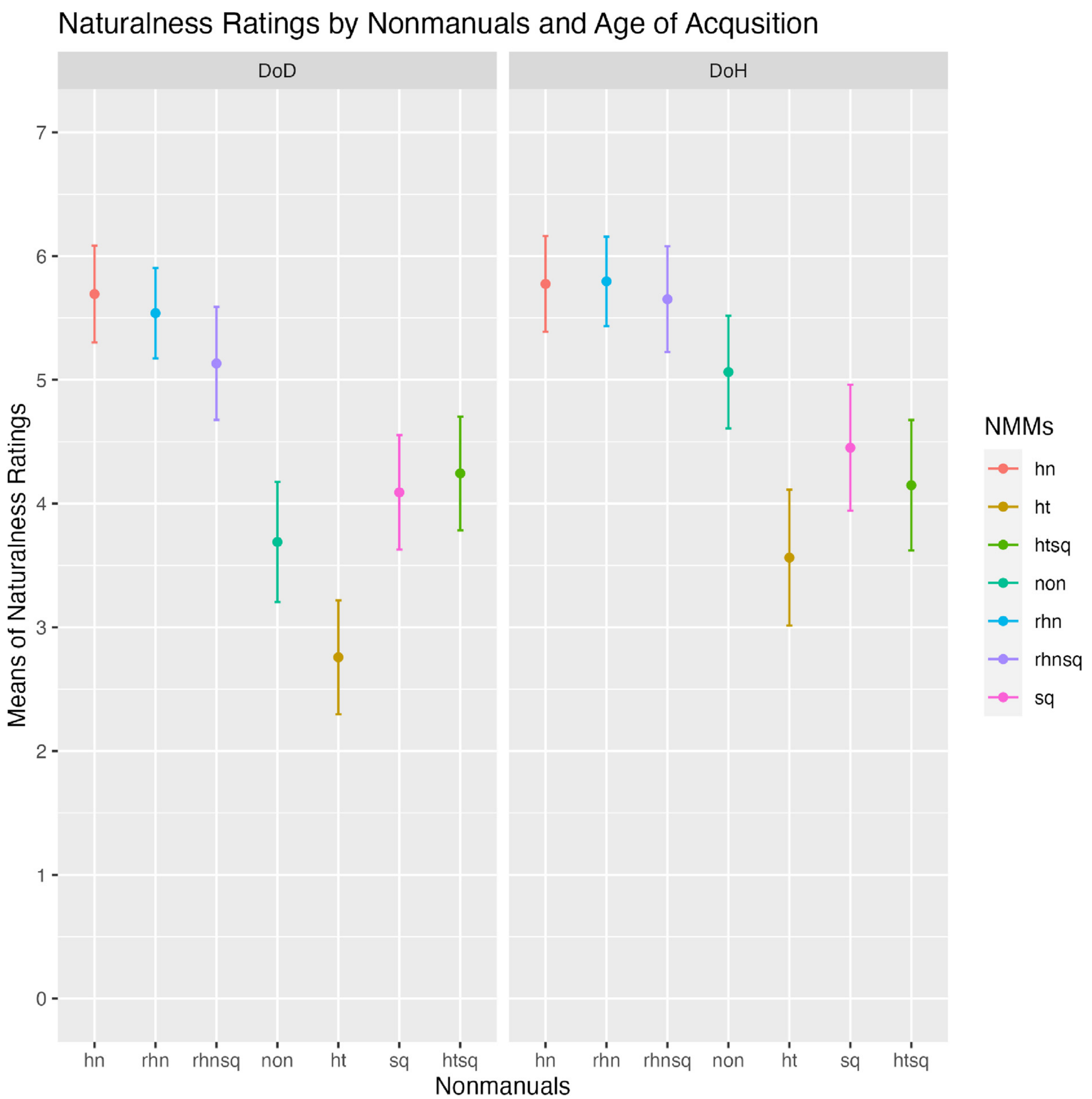

Fig. 8. Effects of nonmanuals and age of acquisition on naturalness ratings.

I will present the results for certainty ratings and certainty attribution in the following sections, but the age of acquisition has an effect only on naturalness ratings. It was not the main effect, but a significant interaction between nonmanuals and age of acquisition did exist. The small participant pool is likely to be one reason for no effect of age of acquisition, and the results need to be replicated with more participants in the future. Bearing in mind this shortcoming, one possible explanation could be the relation between the age of exposure to the language and the performance on metalinguistic tasks. Although their study was not exclusively on nonmanuals, Malaia et al. (2020) similarly investigated native signers, early learners, and late learners' grammaticality judgments and electrophysiological reactions across domains. One of the domains was topic sentences, which include nonmanual markers in Austrian Sign Language (ÖGS). They found that there was a strong positive correlation between the late age of acquisition and higher grammaticality judgments and a strong negative correlation between the age of acquisition and the mean amplitude of N400 response. ÖGS signers with late exposure to sign language did not realize the semantic and syntactic anomalies in the sentences and were generous in their ratings. This may be due to the nonmanuals or the sentence structure in their case, but TİD data does parallel with their findings. Specifically, TİD signers with late exposure to sign language also had higher ratings for nonmanual conditions compared to their peers. In a crosslinguistic study on role-shift, Aristodemo et al. (2023) also posited that poor performance by late Catalan Sign Language (LSC) learners may be because they missed the subtle nonmanual cues in role-shift structures. Considering estimated marginal means comparisons, TİD late signers did not make fine distinctions among conditions as much as their early peers. Lastly, TİD late signers were also found to produce more nonmanuals in focus structures compared to their early peers (Karabüklü and Gürer, 2024). Overall, late signers differ from early signers in their production of nonmanuals and their judgments on them.

4.2. Certainty ratings

A linear mixed effects model was fit to assess the fixed effects of sentences, nonmanuals, and age of acquisition on the certainty ratings. A series of models was also run to test the effects of confounding factors like gender and age. None of these factors had significant effects on the certainty ratings. There was no significant effect of the age of acquisition on certainty rating. Model comparisons showed significant main effects of sentences ($X^2(11) = 36.795$, $p < 0.001$) and nonmanuals ($X^2(6) = 134.15$, $p < 0.001$) on certainty ratings. There was a significant interaction of sentences and nonmanuals ($X^2(6) = 87.393$, $p = 0.04$). Table 7 presents random and fixed effects, the model likelihood, AIC and BIC in the full model.

Table 7
Certainty model summary and fit statistics.

Random effects	**Variance**		**SD**	
Item	0.004		0.07	
Participant	0.71		0.84	
Number of obs: 1372	Groups: Trial = 176		Participant = 15	
Fixed Effects	***df***	***F***	***p***	
Sentence	11	36.795	<0.001	
NMM	6	134.15	<0.001	
Sentence*NMM	66	87.393	0.04	
Deviance (REML)	**log Lik.**	***df***	**AIC**	**BIC**
5460.2	−2680.1	1285	5524.2	5988.7

4.2.1. Effects of sentences

Pairwise comparisons of estimated marginal means of sentence types showed that the signer in the sentences with OLUMLU (positive) was rated significantly more certain than the ones with BIL (know) ($\beta = 1.22$, $SE = 0.34$, $t = 3.61$, $p = 0.02$), LAZIM (necessary) ($\beta = 1.33$, $SE = 0.33$, $t = 4.05$, $p = 0.004$), MECBUR (obligatory) ($\beta = 1.39$, $SE = 0.33$, $t = 4.23$, $p = 0.002$), OLABILIR (possible) ($\beta = 1.54$, $SE = 0.31$, $t = 4.95$, $p < 0.001$), OL LAZIM (be necessary) ($\beta = 1.45$, $SE = 0.33$, $t = 4.42$, $p = 0.001$), and declarative sentences ($\beta = 1.17$, $SE = 0.33$, $t = 3.55$, $p = 0.02$).

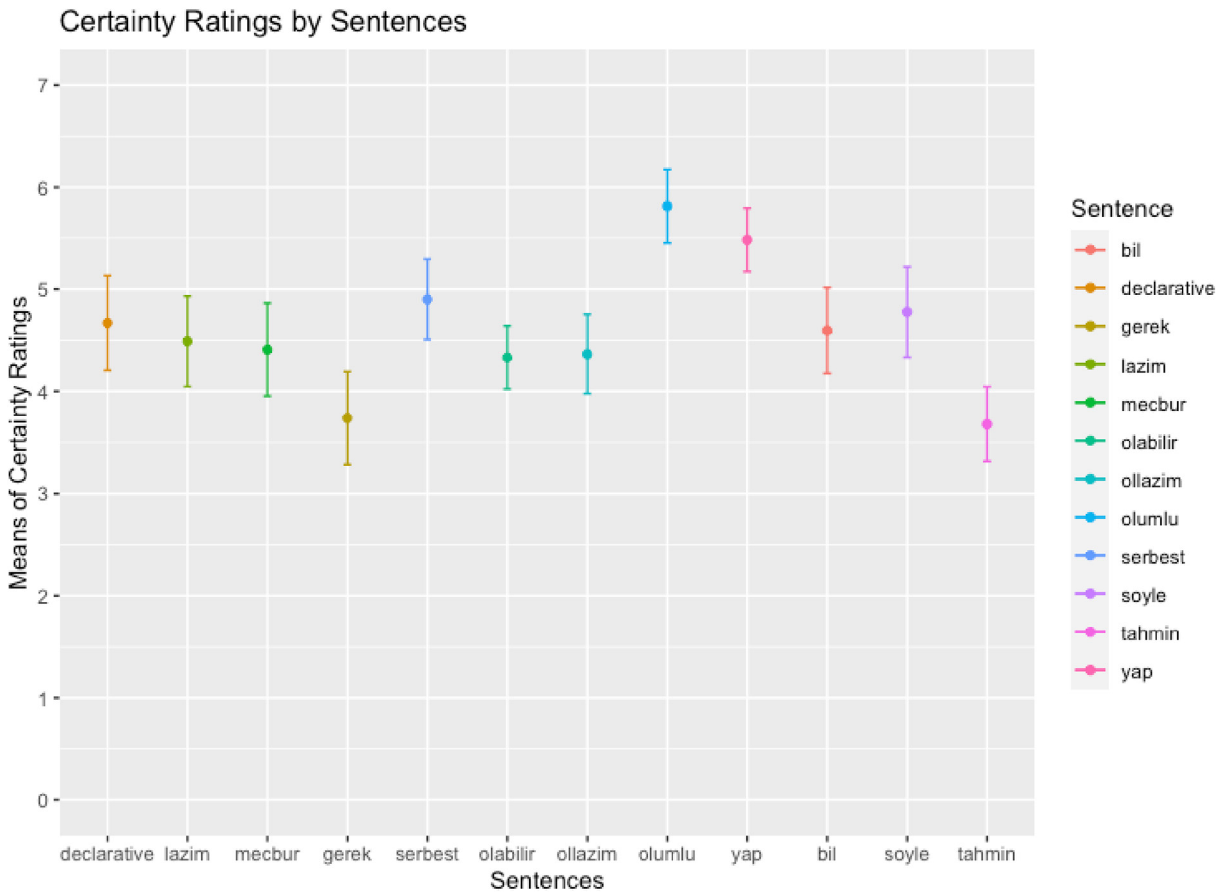

Fig. 9. Effects of sentence types on certainty ratings.

As seen in Fig. 9, participants found the signer significantly less certain in the sentences with TAHMIN (guess) than OLUMLU (positive) ($\beta = -2.20$, $SE = 0.34$, $t = -6.42$, $p < 0.001$), SERBEST (free) ($\beta = -1.29$, $SE = 0.34$, $t = -3.76$, $p = 0.01$), SOYLE (tell) ($\beta = -1.17$, $SE = 0.34$, $t = -3.40$, $p = 0.04$), and YAP (do) ($\beta = -1.41$, $SE = 0.33$, $t = -4.28$, $p = 0.001$).

Lastly, the signer with GEREK (required) was found significantly less certain than OLUMLU (positive) ($\beta = -2.09$, $SE = 0.33$, $t = -6.35$, $p < 0.001$), SERBEST (free) ($\beta = -1.17$, $SE = 0.33$, $t = -3.57$, $p = 0.02$), and YAP (do) ($\beta = -1.29$, $SE = 0.31$, $t = -4.11$, $p = 0.002$). As naturalness ratings showed, participants found GEREK significantly less natural. It was also reported in the literature that TİD signers found GEREK as non-native TİD sign which was borrowed from Turkish (Karabüklü, 2022). These might be why the certainty ratings of GEREK were affected by its naturalness and lexical status.

4.2.2. Effects of nonmanuals

Across all sentence types, head nod, repetitive head nod, and repetitive head nod and squint increased the certainty ratings compared to no nonmanual condition while head tilt, squint, and head tilt and squint decreased the certainty ratings (Fig. 10). Estimated marginal means comparisons showed that participants found the signer significantly more certain in the sentences with head nod compared to head tilt ($\beta = 2.39$, $SE = 0.25$, $t = 9.41$, $p < 0.001$), squint ($\beta = 1.42$, $SE = 0.25$, $t = 5.81$, $p < 0.001$), head tilt and squint ($\beta = 2.11$, $SE = 0.25$, $t = 8.47$, $p < 0.001$), and no nonmanual condition ($\beta = 1.11$, $SE = 0.25$, $t = 4.46$, $p < 0.001$). Similarly, conditions with repetitive head nod yielded higher certainty levels. Participants found the signer significantly more certain in the sentences with head nod compared to the ones with squint ($\beta = 1.31$, $SE = 0.24$, $t = 5.47$, $p < 0.001$). Participants also found the signer significantly more certain in the conditions with repetitive head nod and squint than no nonmanual conditions ($\beta = 0.76$, $SE = 0.25$, $t = 3.03$, $p = 0.04$).

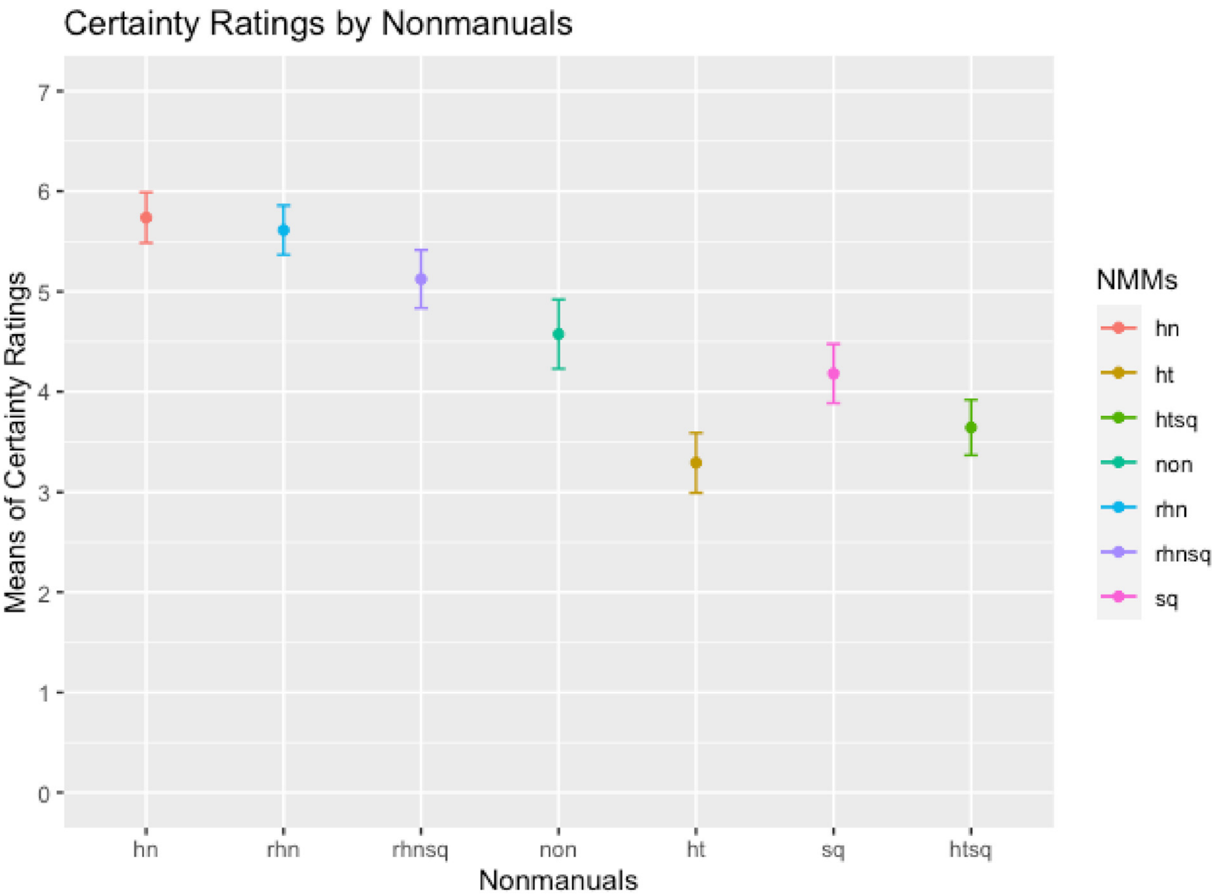

Fig. 10. Effects of nonmanuals on certainty ratings.

As seen in Fig. 10, participants found the signer less certain in the conditions with head tilt and squint. More specifically, they rated the conditions with head tilt significantly less certain than the conditions with repetitive head nod ($\beta = -2.28$, $SE = 0.25$, $t = -9.16$, $p < 0.001$), squint ($\beta = -0.97$, $SE = 0.25$, $t = -3.83$, $p = 0.003$), repetitive head nod and squint ($\beta = -1.73$, $SE = 0.26$, $t = -6.64$, $p < 0.001$), and no nonmanual ($\beta = -1.28$, $SE = 0.26$, $t = -5.00$, $p < 0.001$). As previously stated, head tilt was the least natural nonmanual as a single nonmanual, thus its low ratings may reflect its naturalness.

The signer in the conditions with head tilt and squint was also rated significantly less certain than the conditions with repetitive head nod ($\beta = -2.00$, $SE = 0.24$, $t = -8.20$, $p < 0.001$), repetitive head nod and squint ($\beta = -1.45$, $SE = 0.26$, $t = -5.66$, $p < 0.001$), and no nonmanual ($\beta = -1.00$, $SE = 0.25$, $t = -3.97$, $p = 0.002$). The results replicated Karabüklü and Wilbur's (2020) findings in a more systematic way across different sentence types. Parallel to their findings, head nod and repetitive head nod yielded higher certainty levels while head tilt and squint yielded lower certainty levels. Moreover, the significant difference between head nod and squint ($\beta = 1.42$, $SE = 0.25$, $t = 5.81$, $p < 0.001$) supports another crucial finding that head nod and squint behaved as polar ends of certainty.

4.2.3. Interaction of sentences and nonmanuals

When the interaction between sentences and nonmanuals was examined the significant effects of nonmanuals were preserved in some sentences while they were lost in others. To illustrate, the effects of nonmanuals were not so prominent in the sentences with the matrix verb TAHMIN (guess) as seen in Fig. 11. In contrast, nonmanuals with declarative sentences yielded more significant distinctions. Table 8 summarizes the significant interactions; the exhaustive list of all contrasts is presented in the Supplementary Materials.

Fig. 11. Effects of nonmanuals and sentences on certainty ratings.

As seen in Table 8, head nod and repetitive head nod largely yielded higher certainty ratings with most sentence types from declarative, epistemic modals OLABILIR (possible), OL LAZIM (be necessary), or permission modal SERBEST (free). Head tilt and squint together is another nonmanual that robustly appears at the other direction on the certainty scale. That is, it yielded less certainty with most sentences like MECBUR (obligatory) or YAP (do). Lastly, some nonmanuals like repetitive head nod and squint yielded significant differences in certain cases where the signer's certainty can be further boosted like in declarative sentences but not in deontic sentences like LAZIM or MECBUR.

While nonmanuals yielded similar effects as increasing the threshold in case of head nod or decreasing the threshold in case of squint, the threshold came from the sentence, in other words, from the manual domain. To illustrate, repetitive head nod increased the certainty threshold across the sentences, yet the certainty level of declarative (bare assertion) with repetitive head nod ($M = 6.67$, $SE = 0.65$) is significantly higher than the certainty level of the sentence with epistemic OL LAZIM (be necessary) and repetitive head nod ($M = 3.80$, $SE = 0.65$) ($\beta = 2.87$, $SE = 0.87$, $t = 3.31$, $p = 0.05$). Similarly, the certainty level of the sentence with ability modal OLUMLU (positive) and repetitive head nod ($M = 6.84$, $SE = 0.65$) is significantly higher than the one with OL LAZIM (be necessary) and repetitive head nod ($\beta = 3.04$, $SE = 0.87$, $t = 3.51$, $p = 0.03$). The certainty level of the sentence with ability modal YAP (do) and repetitive head nod ($M = 6.17$, $SE = 0.43$) is also significantly higher than the one with OL LAZIM (be necessary) and repetitive head nod ($\beta = 2.37$, $SE = 0.71$, $t = 3.33$, $p = 0.04$). In terms of a pattern, squint decreased the certainty threshold across the sentences, yet the threshold of the sentence with ability modal OLUMLU (positive) and squint ($M = 5.99$, $SE = 0.65$) was significantly higher than the one with attitude verb TAHMIN (guess) and squint ($M = 3.20$, $SE = 0.55$) ($\beta = 2.79$, $SE = 0.79$, $t = 3.51$, $p = 0.02$). Significant differences among the certainty levels of the same nonmanual with different sentences further supports threshold semantics perspective on the interaction of sentences and nonmanuals by clearly showing the nature of nonmanuals as scalar modifiers.

Table 8
Significant estimated marginal mean comparisons of sentence and nonmanual interactions in certainty ratings.

Sentences	Nonmanuals	β	SE	t	p
Declarative	hn-ht	3.51	0.87	4.05	0.001
	hn-htsq	2.81	0.87	3.24	0.02
	ht-rhn	−3.93	0.87	−4.54	0.0002
	htsq-rhn	−3.23	0.87	−3.72	0.004
	rhn-rhnsq	2.67	0.87	3.09	0.04
BIL	htsq-non	−2.36	0.79	−2.99	0.05
	htsq-rhn	−2.67	0.79	−3.38	0.01
LAZIM	ht-rhn	−2.83	0.87	−3.26	0.02
MECBUR	hn-ht	3.15	0.87	3.64	0.01
	hn-htsq	3.02	0.87	3.46	0.01
	ht-rhn	−2.98	0.87	−3.41	0.01
	htsq-rhn	−2.85	0.88	−3.24	0.02
OLABILIR	hn-ht	2.89	0.87	3.33	0.02
	ht-rhn	−3.07	0.78	−3.96	0.002
	ht-rhnsq	−2.68	0.87	−3.09	0.03
OL LAZIM	hn-ht	2.67	0.87	3.08	0.04
SERBEST	hn-ht	2.57	0.87	2.97	0.05
	ht-rhn	−2.55	0.87	−2.95	0.05
	htsq-rhn	−3.21	0.87	−3.71	0.004
YAP	hn-ht	2.59	0.78	3.34	0.02
	hn-htsq	3.35	0.78	4.32	0.0004
	hn-non	2.40	0.78	3.09	0.03
	ht-rhn	−2.18	0.71	−3.06	0.04
	htsq-rhn	−2.94	0.71	−4.12	0.001
	htsq-rhnsq	−3.11	0.83	−3.73	0.004

4.3. Certainty attribution

Similar to previous analyses, the proportion of choices for certainty attribution was analyzed with a linear mixed effects model. Participants and items were treated as random factors and sentences, nonmanuals, age of acquisition, age, and gender were treated as fixed factors. Among all fixed factors, only sentences had a significant main effect on the certainty attribution ($X^2(11) = 22.779$, $p = 0.02$). Yet, the estimated marginal mean comparisons showed the significant effect of some nonmanuals; thus, nonmanuals were kept as a fixed effect in the final model. Estimated marginal mean comparisons showed that overall, participants attributed the certainty significantly more to the singer than the subject ($\beta = 0.08$, $SE = 0.02$, $z = 4.35$, $p < 0.001$). Thus, who the certainty was attributed to (WhoCertain) was also modeled as a fixed factor and its effect and interaction with sentences and nonmanuals were tested. Table 9 shows the model summary and fit statistics.

Table 9
Certainty attribution model summary and fit statistics.

Random effects	**Variance**		**SD**	
Item	<0.001		<0.001	
Participant	0.001		0.03	
Number of obs: 3048	Groups: Trial = 176		Participant = 15	
Fixed Effects	***df***	***F***	***p***	
Sentence	11	22.779	0.02	
NMM	6	0.258	0.99	
WhoCertain	2	93.448	<0.001	
Sentence*WhoCertain	13	252.38	<0.001	
NMM*WhoCertain	18	22.106	0.2	
Deviance (REML)	**log Lik.**	***df***	**AIC**	**BIC**
4026.3	−2013.2	3000	4122.3	4411.4

4.3.1. Effects of sentences

Estimated marginal mean comparisons showed that the signer was chosen significantly more over the subject in non-embedded sentences like declarative or modals ($p = 0.001$) as in (12). In the embedded sentences with a matrix verb like BIL (know) or TAHMIN (tahmin) (13), the participants attributed the certainty significantly more to the subject of the matrix clause rather than either the signer or the embedded subject ($p < 0.001$). All comparisons are presented in the Supplementary Material. Fig. 12 clearly shows the switched pattern between the signer and the subject based on sentence types.

(12) ALI SWIM OLABILIR
'Ali might be swimming.'

(13) MERVE$_a$ TAHMIN ALI$_b$ SWIM OLABILIR
'Merve thinks that Ali might be swimming.'

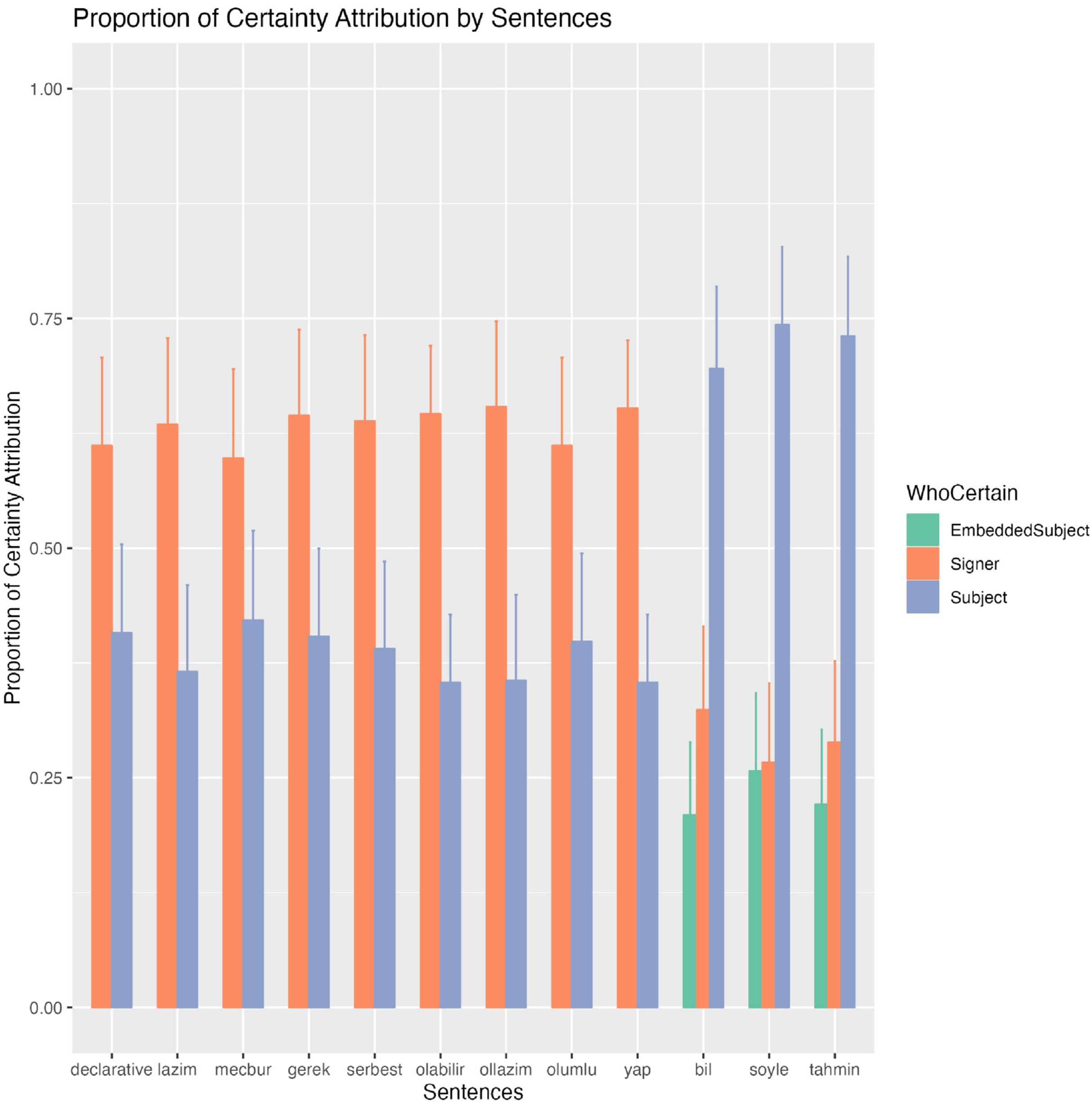

Fig. 12. Proportions of certainty attribution by sentences.

The reason for the switch is that non-embedded sentences are argued to be the reports of the speaker or the signer while embedded sentences are the reports of the subject about the reported proposition (Hintikka, 1969). For example, when a signer uses a sentence as in (12) they report their own inference or knowledge about the possibility of the proposition 'Ali swims'. In contrast, when the signer uses a sentence as in (13), they report Merve's attitude, TAHMIN (guess) in this case refers to the possibility of the proposition 'Ali swims'. While the reporter is the signer in both cases, whose attitude is under discussion shifts from the signer in the case of epistemics (12) to the matrix subject in the case of attitude verbs (13). Results empirically support the theoretical proposal that attitude verbs report the subject's attitude about the embedded proposition.

4.3.2. Effects of nonmanuals

Although nonmanuals did not have a main effect on the certainty attribution, estimated marginal means comparisons showed significant effects of some nonmanuals. Participants attributed the certainty significantly more to the signer than the subject ($\beta = 0.15$, $SE = 0.05$, $z = 3.21$, $p = 0.004$) when the sentences were presented with squint. Similarly, participants attributed the certainty significantly more to the signer when squint occurred with other nonmanuals like repetitive head nod and squint ($\beta = 0.17$, $SE = 0.05$, $z = 3.37$, $p = 0.002$), or head tilt and squint ($\beta = 0.18$, $SE = 0.05$, $z = 3.84$, $p = 0.001$). As seen in Fig. 13, since there was no significant effect of head nod or head tilt when they appeared alone, I propose that it is still the effect of squint when it occurs with other nonmanuals.

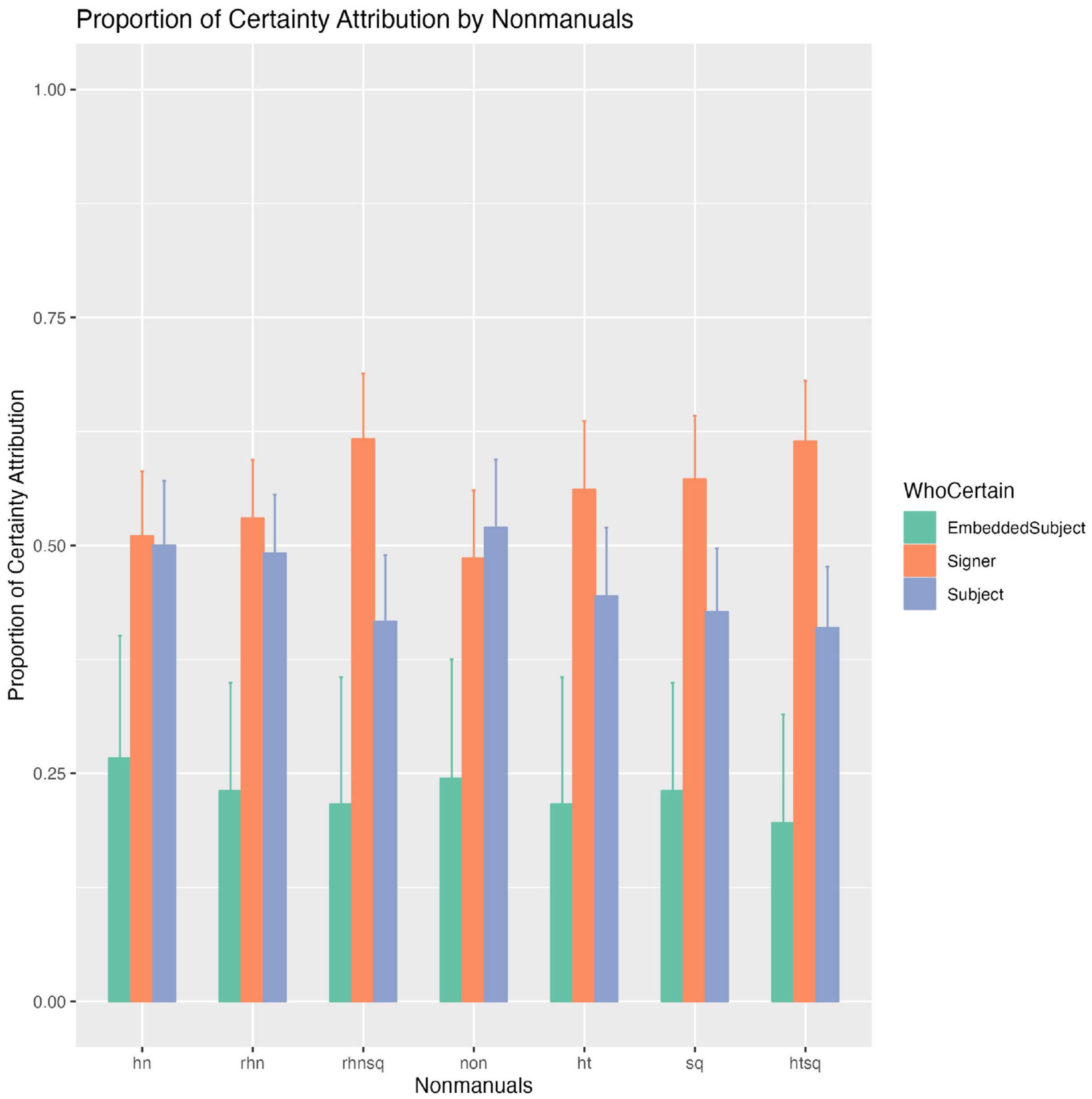

Fig. 13. Proportions of certainty attribution by nonmanuals.

5. Discussion

Signer certainty was affected by both sentence types and nonmanuals. Signer was found to be less certain with epistemic modals and attitude verb TAHMIN (guess). Following Grice (1975), a collaborative signer is expected to convey if they have the best grounds and available evidence for the proposition that she puts forward. Regarding this, the effects of sentences on certainty ratings are expected. If a signer is certain about the truth of the proposition she is expected to use a declarative (bare assertion). Otherwise, she is expected to convey the lack of certainty or the justification for the truth (Krifka, 2023; Partee, 2004) via epistemic, attitude verbs, or other available tools in the language.

Although the effects and the usages of lower degrees of certainty as hedging have been frequently discussed, the effects and usages of higher degrees of certainty are less studied. Scholars had different accounts for the usages of *know* and bare assertion, and their epistemic strengths. Some scholars argued that *know* is epistemically weaker than bare assertion since it can be used in diverse contexts which yield a different range of epistemic standards. In contrast, some scholars argued that speakers would use *know* in high-certainty situations since it has higher epistemic standards than bare assertions (DeRose, 2002). Empirical findings showed that speakers used bare assertions in situations when they were highly certain about the proposition (Schuster and Dengen, 2020) or when the proposition was unequivocally true (Turri, 2013). Contributing to these findings, results showed that participants rated bare assertions (declarative) ($M = 4.67$, $sd = 2.17$) at the same certainty level as the ones with attitude verb BIL (know) ($M = 4.59$, $sd = 2.17$). In spoken languages, the epistemic strengths of attitude verbs and bare assertion were shown to be affected and re-ordered in different social goals as briefing or interrogating (Lorson et al., 2021). In the case of TID, the dynamicity of bare assertion and attitude verbs become obvious in their interactions with nonmanuals. In other words, their epistemic strengths changed based on which nonmanual with which they appeared. To illustrate, declarative ($M = 4.86$, $sd = 2.50$) had a lower certainty level than BIL ($M = 5.59$, $sd = 1.79$) with no nonmanual. Declarative had a higher certainty level ($M = 6.25$, $sd = 1.74$) than BIL ($M = 5.55$, $sd = 1.76$) with head nod while it had a lower certainty level ($M = 4.00$, $sd = 2.43$) than BIL ($M = 4.69$, $sd = 1.98$) with repetitive head nod and squint.

As the first systematic study testing the effects of nonmanuals, the current study also showed that nonmanuals yielded distinct certainty effects. When we examine the results, there are clearly two groups: those that increase the certainty and those that decrease it. Signer certainty was boosted with head nod and repetitive head nod while it decreased with head tilt and squint. I will not consider head tilt as a separate factor since it was found unnatural alone, but natural with squint. Thus, it may be a secondary cue and requires further examination. I will consider repetitive head nod as a different realization rather than a distinct marker. Of head nod and squint, the former usually yielded higher ratings while the latter yielded lower ratings. Furthermore, Karabüklü and Wilbur (2020) reported that a sharp head nod and squint did not cooccur in their production data with the same certainty effects and the results here replicated their findings. We know the semantic roles of modals or attitude verbs but not the functions of nonmanuals. They may be the certainty markers, yet it is clear that the certainty level was affected by not only nonmanuals but also sentences. Another option is that certainty is the effect of the semantic computation of sentences and nonmanuals. I will review the possibilities for their functions in the following subsection.

5.1. Why are there two exponents of certainty in nonmanual channel?

Considering all the findings and the effects, the following question arises: Why are there two distinct exponents for the same notion? In other words, is head nod a certainty marker while squint is an uncertainty marker? At an initial glance, having distinct morphemes (i.e., head nod and squint) from two distinct articulators (head and eyes) seems to be redundant. That is, it is possible to have only head or eyes and take neutral position or null morpheme (no head nod or no squint) as one end of the scale. In the case of head nod, no head nod would be lower end of the certainty scale, and realizations of head (slight, repetitive, or sharp) nod could appear with the increased degree. In a similar fashion for squint, no squint would be the upper end of the certainty scale, and realizations of squint could appear with the decreased degree.

Either TİD has two exponents for different parts of the same scale and their modifications, or head nod and squint have distinct functions other than certainty, and the effects observed in the results could be an interaction. As for head nod, this option might be the focus marking. In other words, certainty level is found to be boosted when focus appears over the verb in spoken languages (Prieto and Roseano, 2021; Tonhauser, 2016). Sign languages are also well-known to mark focus with nonmanuals (Wilbur, 2012; Kimmelman and Pfau, 2021), thus high certainty level may be the effect of verb focus. Yet, a recent experimental study investigated how focus was realized in manual and nonmanual domains in a production design (Karabüklü and Gürer, 2024). Authors found that focus is marked with a longer duration in the manual domain and focus did not have a significant effect on nonmanual production. They mostly observed head nod (72% among all nonmanuals) in their dataset, yet overall nonmanuals appeared only in 22% of their data. Based on their results, head nod was not a focus marker in TİD, yet the current data revealed a high certainty when it appeared over the verb or modal. Since some languages distinguish verum focus or polarity focus from verb focus, head nod may still be a verum focus marker or polarity focus marker that did not appear in their data. Furthermore, the negative polarity is always marked with head back tilt (hbt) in TİD (Gökgöz, 2011) and it overrides head nod (hn). One study reported head nod over the sign YES and at the end of the sentence (Açan Aydın, 2014). Although the author did not report whether these sentences (14) and (15) were responses to polar questions, sentence (14) includes the sign YES accompanied by head nod. Interestingly, sentence (15) does not include the manual sign, but starts with the head nod and mouthing of the manual sign 'yes'. Açan Aydın (2014) proposes that head nod in these sentences might function ‘to mark the end of a sentence or a clause, to strengthen the affirmation in declarative sentences, or to stress the focus in the sentence’ (p. 88). Bearing in mind Karabüklü and Gürer's (2024) findings on focus marking, the data from Açan Aydın (2014) suggest that head nod might be the emphasis on positive polarity. Yet, this proposal needs in-depth future investigation.

(14)
hn   hn   hn
YES MUSIC/SONG VIOLIN PLAY CHILD STUDENT existential
‘Yes, there are school children who are playing music and violins.’
(TİD Açan Aydın 2014: p. 88)

(15)
hn, mouthing (yes)   hn
YAP, NAUGHTY YAP
‘Yes, (he/she) does. (He/she) acts up.’ (TİD Açan Aydın 2014: p. 88)

As for squint, we need to investigate if it yields similar effects with different structures to understand its function. In the literature, squint was observed with attitude verb TAHMIN (guess) (Göksel and Kelepir, 2016), relative clauses (Kubus, 2016), and negative values of a property in comparatives like *thin* or *small* (Özsoy and Kaşıkara, 2017). As shown in the study, squint is not only restricted the verb TAHMIN but it can also appear with other attitude verbs or bare assertion (declarative). Considering all these patterns, there are two possible explanations for squint's function: (i) signaling uncertainty and/or requiring the interlocutor's confirmation, or (ii) being a scalar modifier which decreases the threshold. As for the first account, the uncertainty conveyed by squint is clearly tied to the signer as also seen in certainty attribution results. One possibility under the first account is that the signer only puts the uncertainty to the common ground. Another possibility is that the signer offers uncertainty or marks lack of certainty but also requests for confirmation from the interlocutor in the sense of Heim and Wiltschko's (2020) analysis on sentence-peripheral particle eh in Canadian English. Additional support for this analysis comes from the relative clause data. Kubus (2016) analyzed most of the relative clauses in his data as re-introduced relative clauses and discussed that its function may behave as a re-introduced topic to remind the interlocutor of the discourse referents. With regard to the analysis of re-introduced topics, the signer is certain about the reference in that case, but not certain whether their interlocutor would remember it. It parallels with eh in this case where the signer requests the confirmation for the reference from the signer. As previously stated, squint is always tied to the signer in the current data, yet the uncertainty is to the interlocutor in the re-introduced relative clauses, rather than coming from the signer's lack of information or evidence. Furthermore, squint was also significantly attributed to the signer more than other options (subject or embedded subject). Either context or another mechanism needs to shift uncertainty to the interlocutor.

The challenge to the first account comes from the occurrence of squint with adjectives and/or comparatives. The signer is certain about the references and presumably the interlocutor is, too. Based on this observation and the interaction of nonmanuals and sentences, another option is that squint behaves as a scalar modifier. Squint did not yield the same certainty level across the sentences, yet it decreased the threshold in a similar vein across the sentences. For example, it did not have a significant effect on the certainty level with the verb TAHMIN, yet yielded significantly lower certainty levels with BIL (know) and bare assertions. Thus, these effects suggest that it behaves as a scalar modifier as long as the threshold set by the sentence or manual sign is further decreased. Since TAHMIN was already so low in the scale of certainty, it was not affected by the modifications by nonmanuals. Yet, this analysis needs to account for the occurrence of squint in relative clauses. In either account degree modifier or uncertainty marker, squint seems to have a unified function that yields similar effects across structures, which needs future in-depth analysis.

5.2. How do manual and nonmanual domains interact?

Due to their modality and simultaneous structure, sign languages can use both manual and nonmanual channels to convey certainty. As shown in the results, both sentences and nonmanuals yield different degrees of certainty. Then, how do they interact? Are they two independent domains or does one serve as the basis? Based on the interaction results and from the threshold semantics perspective, I propose that the manual domain set the threshold and nonmanuals boost or deboost the threshold.

Although nonmanuals increased or decreased the certainty level, their effects were not significant for all sentences. Their overall effects were the same as head nod increased the certainty in all cases and squint decreased it, yet these effects were not always significant. More specifically, the certainty level of the attitude verb TAHMIN (guess) was not significantly increased or decreased by the nonmanuals. There are two patterns in the data. First, head nod was crucial in boosting the certainty levels of epistemic signs (Table 8). The certainty level of epistemic sign OL LAZIM (be necessary) was significantly increased only by head nod. The certainty level of another epistemic sign OLABILIR (possible) was significantly increased by head nod and repetitive head nod. Second, in contrast, declarative (bare assertion) and the attitude verb BIL (know) were affected by both head nod and squint, thus their certainty levels were either boosted or deboosted based on the nonmanual. For instance, the certainty level of declarative sentences was increased with head nod or repetitive head nod, and decreased with heat tilt and squint, or repetitive head nod and squint (Table 8). The certainty level of the attitude verb BIL (know) was increased with repetitive head nod or decreased with head tilt and squint. These interactions also show that bare assertions (declarative) and BIL have a gradable nature in line with Lorson et al.'s (2021) discussions on *know*.

When these interactions are considered in the framework of threshold semantics, expressions used in sentences (bare assertion, modals, and attitude verbs) should come with a threshold on the scale of certainty. The threshold is either increased or decreased by the nonmanuals based on how high or low the threshold is. For instance, head nod or repetitive head nod increases the threshold while squint and head tilt decrease it. Then, the proposition is true if the probability of the event exceeds that threshold. In our case, if we assume that epistemic sign OLABILIR (possible) in sentence (16) has the threshold 0.5, then the probability of event 'Zeynep is driving.' is true when the event exceeds the threshold 0.5 set by OLABILIR. The threshold would be less than 0.5, maybe 0.4 in the case where a signer uses the same sentence with squint as in sentence (17). The threshold would be more than 0.5, maybe 0.6 or 0.7 in the case where the signer uses head nod instead of squint or no nonmanual as in sentence (18).

(16) ZEYNEP DRIVE OLABILIR
'Zeynep might be driving.'

(17) $\overline{\text{ZEYNEP DRIVE OLABILIR}}^{\text{sq}}$
'Zeynep might be driving.'

(18) ZEYNEP DRIVE $\overline{\text{OLABILIR}}^{\text{hn}}$
'Zeynep might be driving.'

Overall, I propose that the core of certainty level comes with sentences and the set threshold is modified by the nonmanuals in TİD. The results align with the reported patterns in the literature where nonmanuals were used to strengthen or soften the proposition. This study further showed the labor division of certainty in both channels. The findings also contribute to the patterns in gesture studies and spoken languages. Gesture studies on certainty showed that two modalities share the same labor and contribute to certainty similarly (Borràs-Comes et al., 2019; Prieto and Roseano, 2021). Yet, we have seen that sign languages, at least TİD, divide the labor to two channels, manual channel setting the baseline for the certainty and nonmanual channel modifying the set threshold. In this sense, the results show similarity to Heim and Wiltschko's (2020) analysis separating sentence-peripheral particle and sentence-final intonations. They propose that sentence-peripheral particle in Canadian English conveys the speaker's commitment while sentence-final intonation conveys the addressee's engagement in the common ground. Following the possible analyses on head nod and squint, different nonmanuals may also target different components of common ground. In other words, head nod might be still in the commitment while squint might be in the engagement if their analyses are on tract. Although these proposals need future investigations, the results here are the initial evidence of the labor division in the simultaneous layout of sign languages.

Findings can be extended to other sign languages and test how two channels are affected crosslinguistically. More specifically, future research may investigate if there is a sign language which encodes certainty in nonmanual domain and which certainty is modified in manual domain. German Sign Language (DGS), as shown in Section 2 has nonmanuals for epistemic without requiring any manual sign (Herrmann, 2013; Bross, 2020), but can also have epistemic adverbs along with the nonmanuals (Bross, 2020). One prediction could be that the pattern in TİD might be the reverse in DGS where the threshold is set by nonmanuals and adverbial signs modulate the threshold. Another possibility for DGS might be that modulations of thresholds might be also actualized in nonmanual channel with the intensification as reported by Herrmann (2013). It is an intriguing crosslinguistic research on how both channels and their modulations are actualized in terms of certainty, and these map onto common ground management.

6. Conclusion

In this paper, I investigated the effects of cues in manual and nonmanual channels on the ratings of signer certainty in TİD. Three tasks in the study included the naturalness rating to investigate whether sentences are expected to appear with nonmanuals or nonmanuals are independent markers, the certainty rating to test how signer certainty is affected by different sentences and nonmanuals, and the certainty attribution task to explore who the participants attribute to the certainty (i.e., the signer or the subject). I found that nonmanuals are neither a lexical nor a structural part of sentences in the stimuli based on the naturalness ratings. Thus, as sign language researchers, we should test nonmanuals as possible independent markers or morphemes. Results of the certainty attribution task showed that signers attribute unembedded sentences to the signer and embedded sentences to the subject. Further support for the independence of nonmanuals came from the certainty attribution of squint to the signer.

Signers use cues in both manual and nonmanual channels for the expression of certainty in TİD. Epistemic strength is encoded with both the expression that the signer chose to use as declarative, modal, or attitude verb, and the nonmanual cue the signer use as head nod or squint. Interaction results show that manual domain set the threshold and the threshold is further modified by nonmanuals, thus showing the dynamicity of thresholds. As the first systematic study on the simultaneous expressions of certainty, these results shed light on the role of nonmanuals in sign languages' pragmatics and how visual modality is organized for the expression of certainty.

Data availability/Supplementary files

Data and analysis script are available in the following link: https://osf.io/pu87y/?view_only=6f47a200b5ca460cb6891db4526f2aa9

Ethics and consent

Purdue University's IRB approvals were obtained.

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CRediT authorship contribution statement

Serpil Karabüklü: Writing – review & editing, Writing – original draft, Visualization, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

Author expresses no competing interests.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

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References

Açan Aydın, Zeynep, 2014. Sign languages and aspects of Turkish Sign Language (TİD). J. Fac. Lett./Edebiyat Fakultesi Dergisi 31, 73–96.

Akahori, Hitomi, Yano, Uiko, Matsuoka, Kazumi, Norie, Oka, 2013. Expressing modality: a descriptive study of Japanese sign language. In: Proceedings of the 147th Meeting of Linguistic Society of Japan. University of Foreign Studies, Kobe, Japan, pp. 1–6.

Aristodemo, Valentina, Giustolisi, Beatrice, Zorzi, Giorgia, Gras, Doriane, Hauser, Charlotte, Sala, Rita, Sánchez Amat, Jordina, Donati, Caterina, Cecchetto, Carlo, 2023. On the nature of role shift: insights from a comprehension study in different populations of LIS, LSC and LSF signers. Nat. Lang. Linguist. Theory 41, 459–500. https://doi.org/10.1007/s11049-022-09539-0.

Barr, Dale J., Levy, Roger, Scheepers, Christoph, Tilly, Harry J., 2013. Random effects structure for confirmatory hypothesis testing: keep it maximal. J. Mem. Lang. 68 (3), 255–278. https://doi.org/10.1016/j.jml.2012.11.001.

Bates, Douglas, Maechler, Martin, Bolker, Ben, Walker, Steven, 2014. lme4: Linear Mixed-effects Models Using Eigen and S4. R package version 1.1-7.

Borràs-Comes, Joan, Evangelia, Kiagia, Prieto, Pilar, 2019. Epistemic intonation and epistemic gesture are mutually co-expressive: empirical results from two intonation-gesture matching tasks. J. Pragmat. 150, 39–52. https://doi.org/10.1016/j.pragma.2019.07.004.

Boyas Braem, Penny, Tissi, Katja, 2023. In your face: im/politeness in signed languages, with examples from Swiss German Sign Language (DSGS). In: Jucker, Andreas H., Brown, Lucien, Hübscher, Iris (Eds.), Multimodal Im/politeness: Signed, Spoken, Written. John Benjamins Publishing Company, Amsterdam, pp. 27–63. https://doi.org/10.1075/pbns.333.

Bross, Fabian, 2020. The clausal syntax of German sign language. a cartographic approach. Lang. Sci. Press. https://doi.org/10.5281/zenodo.3560718.

DeRose, Keith, 2002. Assertion, knowledge, and context. Phil. Rev. 111 (2), 167–203. https://doi.org/10.2307/3182618.

Gökgöz, Kadir, 2011. Negation in Turkish Sign Language: the syntax of nonmanual markers. Sign Lang. Linguist. 14 (1), 49–75. https://doi.org/10.1075/bct.53.04gok.

Göksel, Aslı, Kelepir, Meltem, 2016. Observations on clausal complementation in Turkish Sign Language. In: Pfau, Roland, Steinbach, Markus, Herrmann, Annika (Eds.), A Matter of Complexity: Subordination in Sign Languages. De Gruyter Mouton, Berlin, Boston, pp. 65–94. https://doi.org/10.1515/9781501503238-005.

Grice, H. Paul, 1975. Logic and conversation. In: Cole, Peter, Morgan, Jerry L. (Eds.), Syntax and Semantics 3: Speech Acts. Brill, Leiden, pp. 41–58. https://doi.org/10.1163/9789004368811_003.

Hacquard, Valentine, 2006. Aspects of Modality. Massachusetts Institute of Technology dissertation.

Heim, Johannes, 2019. Commitment and Engagement: the Role of Intonation in Deriving Speech Acts. The University of British Columbia. https://doi.org/10.14288/1.0380772 dissertation.

Heim, Johannes, Wiltschko, Martina, 2020. Interaction at the syntax–prosody interface. In: Kentner, Gerrit, Kremers, Joost (Eds.), Prosody in Syntactic Encoding. De Gruyter, Berlin, Boston, pp. 189–218. https://doi.org/10.1515/9783110650532-007.

Heritage, John, 2013. Action formation and its epistemic (and other) backgrounds. Discourse Stud. 15, 551–578. https://doi.org/10.1177/1461445613501449.

Herrmann, Annika, 2013. Modal and Focus Particles in Sign Languages: A Cross-Linguistic Study. De Gruyter Mouton. https://doi.org/10.1515/9781614511816.

Hintikka, Jaakko, 1969. Semantics for propositional attitudes. In: Davis, J.W., Hockney, D. (Eds.), Philosophical Logic. D. Reidel Publishing Company, Dordrecht, Holland, pp. 189–218. https://doi.org/10.1007/978-94-010-1711-4_6.

Holtgraves, Thomas, Perdew, Audrey, 2016. Politeness and the communication of uncertainty. Cognition 1–10. https://doi.org/10.1016/j.cognition.2016.05.005.

Hrastinski, Iva, Wilbur, Ronnie B., 2016. Academic achievement of deaf and hard of hearing children in an ASL/English bilingual program. J. Deaf Stud. Deaf Educ. 21, 156–170. https://doi.org/10.1093/deafed/env072.

Karabüklü, Serpil, Gürer, Aslı, 2024. Prosody of focus in Turkish Sign Language. Language and Cognition Journal 1–34. https://doi.org/10.1017/langcog.2024.4.

Karabüklü, Serpil, 2022. Modal signs and cooccurring nonmanual markers in Turkish Sign Language (TİD): Purdue University dissertation. https://doi.org/10.25394/PGS.21325785.v1.

Karabüklü, Serpil, Wilbur, Ronnie B., 2020. Seeing possible worlds. In Proceedings of Chicago Linguistic Society 55th Meeting. 189–204.

Karabüklü, Serpil, Bross, Fabian, Wilbur, Ronnie B., Hole, Daniel, 2018. Modal signs and scope relations in TİD. In Proceedings of Formal and Experimental Advances in Sign Language Theory (FEAST). 82–92. https://doi.org/10.31009/FEAST.i2.07.

Kimmelman, Vadim, Pfau, Roland, 2021. Information structure: theoretical perspectives. In: Quer, Josep, Pfau, Roland, Herrmann, Annika (Eds.), The Routledge Handbook of Theoretical and Experimental Sign Language Research. Routledge, New York, pp. 391–413.

Krifka, Manfred, 2017. Epistemic, evidential and discourse modalities in commitment space semantics. In: Handout from Workshop Not-at-issue-Meaning and Information Structure. University of Oslo.

Krifka, Manfred, 2023. Layers of assertive clauses: propositions, judgments, commitments, acts. In: Hartmann, Jutta M., Wölstein, Angelika (Eds.), Propositionale argumente im sprachvergleich: Theorie und empirie [Propositional Arguments in Cross-linguistic Research: Theoretical and Empirical Issues]. Narr, Tübingen, pp. 391–413. https://doi.org/10.24053/9783823394105.

Kubus, Okan, 2016. Relative Clause Constructions in Turkish Sign Language. Hamburg University dissertation.

Lassiter, Daniel, 2017. Graded Modality: Qualitative and Quantitative Perspectives. Oxford University Press.

Lenth, Russell, 2019. Emmeans: estimated marginal means, aka least-squares means.

Lillo-Martin, Diane, Smith, Neil, Tsimpli, Ianthi, 2020. Age of acquisition effects in language development. In: Morgan, Gary (Ed.), Understanding Deafness, Language and Cognitive Development. Essays in Honour of Bencie Woll. John Benjamins Publishing Company, Amsterdam, pp. 93–114. https://doi.org/10.1075/tilar.25.06lil.

Lorson, Alexandra, Cummins, Chris, Rohde, Hannah, 2021. Strategic use of (un)certainty expressions. Front. Commun., 635156 https://doi.org/10.3389/fcomm.2021.635156.

Lorson, Alexandra, Rohde, Hannah, Cummins, Chris, 2023. Epistemicity and communicative strategies. Discourse Process 556–593. https://doi.org/10.1080/0163853X.2023.2255494.

Malaia, Evie, Krebs, Julia, Roehm, Dietmar, Wilbur, Ronnie B., 2020. Age of acquisition effects differ across linguistic domains in sign language: Eeg evidence. Brain Lang. 1–10. https://doi.org/10.1016/j.bandl.2019.104708.

Mayberry, Rachel I., 2007. When timing is everything: age of first-language acquisition effects on second-language learning. Brain Lang. 537–549. https://doi.org/10.1017/S0142716407070294.

Mayberry, Rachel I., Eichen, Ellen B., 1991. The long-lasting advantage of learning sign language in childhood: another look at the critical period for language acquisition. J. Mem. Lang. 486–512. https://doi.org/10.1016/0749-596X(91)90018-F.

Mayberry, Rachel I., Kluender, Robert, 2018. Rethinking the critical period for language: new insights into an old question from american sign language. Biling. Lang. Cognit. 486–512. https://doi.org/10.1017/S1366728918000585.

Mayberry, Rachel I., Lock, Elizabeth, Kazmi, Hena, 2002. Linguistic ability and early language exposure. Nature 38. https://doi.org/10.1038/417038a.

Moya-Avilés, Berta, Barberà, Gemma, Bach, Carme, 2023. Politeness in Catalan Sign Language (LSC). In: Jucker, Andreas H., Brown, Lucien, Hübscher, Iris (Eds.), Multimodal Im/politeness: Signed, Spoken, Written. John Benjamins Publishing Company, Amsterdam, pp. 65–98. https://doi.org/10.1075/pbns.333.

Özsoy, A Sumru, Kaşıkara, Hüner, 2017. Comparatives in Turkish Sign Language (TİD). Linguist. Discov. Spec. Ed. 47–63. https://doi.org/10.1349/PS1.1537-0852.A.485.

Özsoy, A Sumru, Kelepir, Meltem, Nuhbalaoğlu, Derya, Hakgüder, Emre, 2018. Properties of command constructions in TİD. Dilbilim Araştırmaları Dergisi (Journal of Linguistics Research) 161–178. https://doi.org/10.18492/dad.373380.

Partee, Barbara, 2004. Comments on jason stanley's 'on the linguistic basis for contextualism'. Phil. Stud. 147–159.

Pfau, Roland, Göksel, Aslı, Hosemann, Jana, 2021. Our Lives - Our Stories: Life Experiences of Elderly Deaf People. De Gruyter Mouton, Berlin. https://doi.org/10.1515/9783110701906.

Prieto, Pilar, Roseano, Paolo, 2021. The encoding of epistemic operations in two romance languages: the interplay between intonation and discourse markers. J. Pragmat. 172, 146–163. https://doi.org/10.1016/j.pragma.2020.11.008.

Ramchand, Gillian, 2014. Stativity and present tense epistemics. In: Todd, Snider, D'Antonio, Sarah, Weigand, Mia (Eds.), Proceedings of Semantics and Linguistic Theory Conference (Salt 24). Linguistic Society of America, pp. 102–121. https://doi.org/10.3765/salt.v24i0.2367.

Schuster, Sebastian, Dengen, Judith, 2020. I know what you're probably going to say: listener adaptation to variable use of uncertainty expressions. Cognition 203, 104285. https://doi.org/10.1016/j.cognition.2020.104285.

Shaffer, Barbara, 2004. Information ordering and speaker subjectivity: modality in ASL. Cognit. Ling. 15, 175–195. https://doi.org/10.1515/cogl.2004.007.

Siyavoshi, Sara, 2019. Hands and faces: the expression of modality in ZEI, Iranian Sign Language. Cognit. Ling. 30, 655–686. https://doi.org/10.1515/cog-2018-0130.

Swerts, Marc, Krahmer, Emial, 2005. Audiovisual prosody and feeling of knowing. J. Mem. Lang. 53, 81–94. https://doi.org/10.1016/j.jml.2005.02.003.

Tonhauser, Judith, 2016. Prosodic cues to presupposition projection. In: Moroney, Mary, Little, Carol-Rose, Collard, Jacob, Burgdorf, Dan (Eds.), Proceedings of Semantics and Linguistic Theory Conference (Salt 26). Linguistic Society of America, pp. 934–960. https://doi.org/10.3765/salt.v26i0.3788.

Tubau, Susagna, González-Fuente, Santiago, Prieto, Pilar, Espinal, Maria Teresa, 2015. Prosody and gesture in the interpretation of yes-answers to negative yes/no-questions. Linguist. Rev. 32, 115–142. https://doi.org/10.1515/tlr-2014-0016.

Turri, John, 2013. The test of truth: an experimental investigation into the norm of assertion. Cognition 129, 279–291. https://doi.org/10.1016/j.cognition.2013.06.012.

Wilbur, Ronnie B., 2000. The use of ASL to support the development of English and literacy. J. Deaf Stud. Deaf Educ. 5, 81–104.

Wilbur, Ronnie B., 2012. Information structure. In: Pfau, Roland, Steinbach, Markus, Woll, Bencie (Eds.), Sign Language: an International Handbook. De Gruyter Mouton, Berlin, pp. 462–489. https://doi.org/10.1515/9783110261325.462.
Yalcin, Seth, 2010. Probability operators. Philos. Compass 5, 916–937. https://doi.org/10.1111/j.1747-9991.2010.00360.x.
Zorzi, Giorgia, Giustolisi, Beatrice, Aristodemo, Valentina, Cecchetto, Carlo, Hauser, Charlotte, Quer, Josep, Sánchez Amat, Jordina, Donati, Caterina, 2022. On the reliability of the notion of native signer and its risks. Front. Psychol. 13, 716554. https://doi.org/10.3389/fpsyg.2022.716554.

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